

The block below carries no module guard, so docstrip copies it into EVERY generated file. That is what we want for the TeX files and wrong for the latexmkrc, which is Perl and does not understand lines opening with two percent signs – hence the negated guard around it.

The guard lines themselves are hidden behind a false conditional. docstrip reads a guard line either way, but ltxdoc, which typesets this file, would otherwise set the literal text of the guard as a stray paragraph in the body of the manual.

Nothing in this area may name a TeX conditional, not even in prose: outside the meta-comment at the top of the file, every documentation line here IS LaTeX source and is executed, and a conditional named in passing opens or closes a real one. That is not hypothetical – an earlier draft of this very comment swallowed the rest of the manual.

Three conditional primitives are deliberately absent from the lists below: the ones that close or branch a conditional. `\DoNotIndex` scans its argument with TeX’s own conditional machinery live, so a bare one of them closes the wrong conditional. The manual ended its run with “Too many }’s” and exited with status 1, which failed the docs step of the harness even though the PDF had been written. Leaving them out of the lists means they get indexed, which is harmless. Note that they cannot be named here either, not even inside `...` shorthand split across a line.

The coppe document class

Version 4.0

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Folha adicional: Coleta CAPES

Since August 2026 every thesis and dissertation must carry a *folha adicional* immediately after the folha de rosto, holding the fields the CAPES/MEC collection requires plus the ficha catalogafica (SiBI manual, 9th rev. ed. 2026, 3.1.2.1.2

and Annex H). The sheet is filled in by the graduate programme, and is neither counted nor numbered.

The macros below feed that sheet. They are institutional data, so the sheet is always typeset in Portuguese regardless of the main language of the work, as the COPPE norm requires of institutional identity.

`\areaconcentracao` The area of concentration, e.g. `\areaconcentracao{Engenharia de Sistemas e Computacao}`. It appears on the folha de rosto, on the folha de aprovacao and on the folha adicional.

`\linhapesquisa` The research line, required by the CAPES model of the folha de rosto.

`\tipoproducao` The kind of intellectual output, one of `bibliografica`, `artistica`, `tecnologica` or `tecnica`. Anything else raises a warning.

`\projetovinculado` Whether the work belongs to a research project (`sim` or `nao`) and, if so, its

`\nomeprojeto` name.

`\agenciafomento` A funding agency, given as full name and acronym; repeat once per agency:

```

\agenciafomento{Conselho Nacional de Desenvolvimento
                  Cientifico e Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a
                  Pesquisa do Estado do Rio de Janeiro}{FAPERJ}

```

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one

`foreignabstract (env.)` page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which

`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is

`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for

`\printlosymbols`

`\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains $\text{\LaTeX}$ commands, mathematical symbols, or mix of uppercase and lowercase. `MakeIndex` needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain `MakeIndex` input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

`COPPE $\text{\TeX}$` produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no `BIB $\text{\TeX}$` `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` `BIB $\text{\TeX}$` styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `La $\text{\TeX}$` , `biber`, and twice again `La $\text{\TeX}$` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic @book (@livro); @article (@artigo); @artigodejornal (newspaper); @incollection (@partedelivro); @inbook (@emlivro); @inreference (@verbete); @proceedings (@anais); @inproceedings (@trabalhodeevento); @thesis (@tese/@dissertacao, with the presets @dissertacaomestrado, @tesedoutorado, @tesephD, @tcc); @report (@relatorio); @periodical (@periodico); @manual; @misc (@diverso); @online.

Special (NBR 6023 §4.2.5–4.2.14) @patent (@patente); @legislation (@legislacao/@atonormat @jurisdiction (@jurisprudencia, with a relator field); @legal (@documentojuridico); @standard (@norma); @music (@partitura); @map (@mapa); @image (@iconografico); @artwork (@obradearte); @video (@audiovisual); @movie (@filme); @letter (@correspondencia); @dataset (@conjuntodedados).

Games and TV @game (@jogo); @boardgame (@jogodetabuleiro); @videogame (@jogoeletronico); @tvshow (@programadetv).

When an entry has no author or editor, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard L^AT_EX command used before all appendices. COPPET_{EX} introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer's *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian babel), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Computer Modern (padrão L^AT_EX)*. Redefine if you loaded `lmodern`, `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPE_TE_X class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.

- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein / \brazilianabstract` API, and how to create a new language pack.

pdfa Produce the deposited file as **PDF/A-2b**, which 2.2(d) of the manual requires of the digital version. The class loads `pdfx` at level `a-2b` and writes the XMP metadata from the document’s own `\title`, `\author`, `\keyword` and `\department`. Level `a-2b` rather than `a-1b`: `a-1b` forbids transparency, which `tcolorbox` and several graphics packages produce. The metadata reaches the PDF on the SECOND run, because `pdfx` reads it from a file the class writes at the end of the first one – which is no burden, since three runs are needed for cross-references anyway.

assinaturas Give every member of the board a signature rule of its own on the folha de aprovacao, as Annex D of the manual shows, instead of the compact list. The spacing adapts to the size of the board.

coorientador List the coadvisors on the three abstract pages, under the advisors, in the same shape the folha de rosto uses. Off by default: the abstract page listing only the advisors is a COPPE tradition, and 3.1.2.1.4 does not require either form.

rascunhoficha While writing, fill the folha adicional with a ficha catalografica typeset by the class itself, so the page is not an empty frame. **Never valid for deposit:** the real ficha comes from the SiBI generator or the Programa’s library, and goes in with `\fichacatalografica`.

twoside, oneside The 2026 edition dropped the verso margins and the deposit is digital, so the class sets one-sided pages with a 3 cm left margin on every folha. **twoside** restores the mirrored layout for anyone who is going to PRINT and bind the work; **oneside** is the default and is there to be explicit.

semmorewrites, morewrites T_EX has sixteen output streams and a work that asks for every list runs out of them; the failure, “No room for a new `\write`”, lands on whichever list came last and explains nothing. Under pdfT_EX the class therefore loads the `morewrites` package by itself when it is installed, which removes the ceiling. **semmorewrites** turns that off, for the rare case where the package disagrees with something else in the document – it buffers what is written until shipout, so a package that writes a file and reads it back in the same run can misbehave. **morewrites** means “I insist”: a missing package becomes an error instead of a warning. Under LuaLaT_EX, which has 128 streams, none of this is loaded and none of it is needed.

numeraisromanos Number the pre-textual folhas in lowercase roman numerals. **Contrary to 2.7**, which has the count running from the folha de rosto with nothing printed until the textual part. Kept only so that documents written under the older norm still compile; do not use it for a new work.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the `abstract` environment. Picked by the class option (`brazilian` by default; one of `english`, `spanish`, `french`, `italian` also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the `foreignabstract` environment. `english` by default for every main language; switches automatically to `brazilian` when `main = english`.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

`\titlein`
`brazilianabstract (env.)`
`\usecoppelanguage`

Two pieces of user-facing API come with the multilingual model:

- `\titlein{<lang>}{<text>}` records the title in any language. `\title{...}` still writes the Brazilian-Portuguese title and `\foreigntitle{...}` still writes the English one (back-compat); for a main language outside the `brazilian/english` pair, also call `\titlein{<lang>}{...}`. The cover and folha de rosto print whichever title matches `\coppe@mainlang`.
- The `brazilianabstract` environment is the optional third abstract used when the main language is neither `brazilian` nor `english` – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in `\begin{otherlanguage}{brazilian}` and uses the Portuguese template strings. A document with `main = brazilian` can simply keep using `abstract`; `brazilianabstract` is redundant then.
- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the `TEXINPUTS` path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for brazilian/english in `coppe.cls` (appendix, annex, frame, listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys available from, urlseen, in, editor, editors, depositor, coppescale, msdiss, dscthis, phdthesis, tcc. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships portuguese, english, spanish, french and italiano – not italian.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T'\i tulo em portugu\~es}      % Brazilian abstract slot
  \foreigntitle{English title}          % English abstract slot
  \titlein{spanish}{T'\i tulo en espa\~nol} % cover/folha
  \author{Juan}{P\~erez}
  \advisor{Prof.}{Mar'\i a}{Garc'\i a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
  \begin{abstract}      Resumen principal en espa\~nol.      \end{abstract}
  \begin{foreignabstract} English foreign abstract.          \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\~on} Cuerpo en espa\~nol\dots
\end{document}
```

6 Files in your project folder

In a normal install where CoppeTeX is part of your `TEXINPUTS` path (e.g. inside the local `texmf` tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}  
\DeclareLanguageMapping{english}{english-coppe}
```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
brazilian (default)	— (built into <code>coppe.cls</code>)
english	— (built into <code>coppe.cls</code>)
spanish	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
french	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
italian	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard `book` class (`12pt`, `a4paper`, `oneside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between `pdfTEX` and the Unicode engines (LuaT_EX/XeT_EX).

`inputenc (utf8)` UTF-8 input; loaded *only under pdfT_EX* (the Unicode engines read UTF-8 natively).

`fontenc (T1)` T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfT_EX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 one-sided margins (3/3/2/2 cm) and header height.
`fancyhdr` running headers and footers.
`titlesec` chapter and section title formatting.
`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.
`tabularx` tables with automatic-width columns.
`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).
`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.
`float` float placement control (e.g. the `[H]` specifier).
`caption` caption typography (the ABNT caption-on-top layout).
`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.
`fancyvrb` extended verbatim.
`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms;
the `natural-language` option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main
and foreign language slots.
`csquotes` context-sensitive quotation marks (the recommended companion of
`biblatex`).
`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles
(`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is
`biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded, ...`).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

`hologo` TeX-family logos (`\hologo{LaTeX}, ...`).

`xcolor` colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdfL^AT_EX will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfL^AT_EX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`**: it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation A quotation longer than three lines is INDENTED BY 4 cm from the left margin – not set with a 4 cm left margin, which is a different and much narrower thing. 2.2(b) of the manual says “recuo de 4 cm da margem esquerda”, so on an A4 page with a 3 cm margin the quoted text starts 7 cm from the edge of the paper. It carries no quotation marks and is set in the smaller font, single spaced. The `coppe` class provides the `longquote` environment, which does all of that; the indentation is `\recuolongquote`, should a Programa ever need to change it.

9 A simple example

```

1 <*example>
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating}           % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz}              % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 \usepackage{unicode-math}
11 %\setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe (v4.0) já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelossymbols
22 \makeloabbreviations
23 % Algumas coisas opcionais que não cabem no coppetex.sty
24 %\renewcommand{\familydefault}{\sfdefault} % para usar fontes SEM serifa no default
25 \renewcommand{\familydefault}{\rmdefault} % para usar fontes COM serifa no default
26
27
28 \begin{document}
29   \title{Titulo da Tese}
30   \foreigntitle{Thesis Title}
31   \author{Nome do Autor}{Sobrenome}

```



```

32 \advisor[UFRJ]{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
33 \advisor[UFRJ]{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
34 % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
35 % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
36 % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
37 % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
38
39 \examiner[UFRJ]{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
40 \examiner[UFRJ]{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
41 \examiner[UFF]{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
42 \examiner[UNICAMP]{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
43 \examiner[USP]{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
44 \department{PESC}
45 \date{06}{2026}
46
47 \keyword{Primeira palavra-chave}
48 \keyword{Segunda palavra-chave}
49 \keyword{Terceira palavra-chave}
50
51 % Dados da folha adicional (Coleta CAPES), obrigatoria desde agosto de 2026.
52 % A folha e preenchida pelo Programa; estes campos apenas a compoem.
53 \areaconcentracao{Engenharia de Sistemas e Computa\c c\~ao}
54 \linhapesquisa{Engenharia de Dados e Conhecimento}
55 \tipoproducao{bibliografica} % bibliografica | artistica | tecnologica | tecnica
56 \projetovinculado{sim} % sim | nao
57 \nomeprojeto{Nome do Projeto de Pesquisa ao qual o trabalho se vincula}
58 \agenciafomento{Conselho Nacional de Desenvolvimento Cientifico e
59 Tecnologico}{CNPq}
60 \agenciafomento{Fundacao Carlos Chagas Filho de Amparo a Pesquisa do
61 Estado do Rio de Janeiro}{FAPERJ}
62 % A ficha oficial vem de fichacatalografica.sibi.ufrj.br; tendo o arquivo,
63 % descomente a linha abaixo e a moldura vazia da folha adicional some.
64 % \fichacatalografica{ficha.pdf}
65
66 \maketitle
67
68 \frontmatter
69 \dedication{A alguém cujo valor é digno desta dedicatória.}
70
71 \chapter*{Agradecimentos}
72
73 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
74 orientadores, colegas, familiares e \as ag\encias de fomento. Lorem ipsum
75 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
76 ut labore et dolore magna aliqua.
77
78 \begin{abstract}
79
80 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
81 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
82 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
83 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
84 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
85 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui

```

```

86 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
87 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
88 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
89 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
90 aspernatur aut odit aut fugit.
91
92 \end{abstract}
93
94 \begin{foreignabstract}
95
96 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
97 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
98 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
99 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
100 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
101 pariat. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
102 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
103 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
104 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
105 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
106 aspernatur aut odit aut fugit.
107
108 \end{foreignabstract}
109
110 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
111 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
112 \listoffigures
113 \listoftables
114 \listofquadtros
115 \listofprogramas
116 \listofalgorithms
117 \printloabbreviations
118 \printlosymbols
119 \tableofcontents
120
121 \mainmatter
122 \chapter{Introdução}
123
124 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
125
126 A classe \verb|coppe| foi criada originalmente por Vicente Helano e George Ainsworth, poré
127 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
128
129 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação
130
131 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.c
132
133 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
134
135 Há um grupo de WhatsApp para perguntas e discussões do estilo: \href{https://github.com/COP
136
137 Bugs devem ser reportados na \href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d
138
139 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p

```

140

141 É importante de notar que essa classe não foi construída sobre a classe `\LaTeX` para a A

142

143 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma

144

145 Este documento não substitui, mas complementa, o documento que descreve a classe.

146

147

148

149 `\chapter{Configurações Iniciais}`

150

151 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c

152

153 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra

154

155 Recomendo ler o documento ‘‘The `\verb*|coppe|` document class’’ para entender melhor todas

156

157 `\section{Linguagem principal do texto}`

158

159 Essa classe considera que o texto principal está em português e algumas partes específicas

160 `\begin{itemize}`

161 `\item` A opção `\verb|english|` deve ser usada no comando `\verb|\documentclass|`.

162 `\item` A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p

163 `\end{itemize}`

164

165 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe `\CopeT`

166

167

168 `\section{Por que usar o \LaTeX}`

169

170 Há uma grande discussão entre usuário de Word e `\LaTeX`, principalmente, quanto ao uso dess

171

172 Nós escolhemos o `\LaTeX` por alguns motivos: grande facilidade de seguir um estilo sem se p

173

174 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob

175

176 `\section{Como e onde usar o \LaTeX}`

177

178 Existem muitos tutoriais de `\LaTeX`, mas basicamente, em 2024, ele é usado em dois ambiente

179 `\begin{enumerate}`

180 `\item` Na sua máquina, instalando uma versão completa como o `\hologo{MiKTeX}`, típico do Win

181 `\item` Usar um ambiente na rede, como o Overleaf.

182 `\end{enumerate}`

183

184 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b

185

186

187 `\chapter{Algumas Regras da COPPE}`

188

189 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei

190

191 É imprescindível definir os símbolos, tal como o

192 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.

193 `\syml{\mathbb{R}}{Conjunto dos números reais}`

194 `\symlbl{$\emptyset}$`{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
195
196 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o `\ve`
197
198 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índices
199
200 `\section{Citações}`
201 `\label{sec:citacoes}`
202
203 Citações curtas podem ser feitas `\enquote`{o comando `quote`} ou direto com “duas crases e do
204
205 `\begin{longquote}`
206 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
207 feita com o ambiente `\verb*|longquote|` The primary objective of this
208 investigation was to determine the feasibility of detecting corrosion in
209 aluminum Naval aircraft components with neutron radiographic interrogation
210 and the use of standard corrosion penetrameters. Secondary objectives
211 included the determination of the effect of object thickness on image quality,
212 the defining of minimum levels of detectability and a preliminary investigation
213 of a means whereby the degree of corrosion could be quantified with neutron
214 radiographic data. `\citep{article-example}`
215 `\end{longquote}`
216
217 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote `\verb*|na`
218
219 Em todo caso, `\textbf{deve}` se tomar enorme atenção com as citações, para evitar ocorrer em
220
221 Quando se cita uma obra por meio de outra (`\textit{apud}`), pela ABNT entra na lista de ref
222
223 `\chapter{Floats}`
224
225 Grande parte dos problemas de iniciantes, e veteranos, em `\LaTeX` é da localização dos `\texti`
226
227 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
228
229 `\textbf{Pela norma atual (manual2024), a legenda}` `\verb|\caption|` `\textbf{de toda ilustração}`
230
231 Quadros são ilustrações cujos dados são delimitados por linhas em `\emph{todas}` as bordas (ao
232
233
234 `\section{Tabelas e Figuras Padrão}`
235
236 Vamos ver uma tabela padrão, como a `\autoref{tab:exemplo_numeros}`.
237
238 `\begin{table}[ht]`
239 `\centering` % Centraliza a tabela
240 `\caption{Exemplo de Tabela de Números}`
241 `\label{tab:exemplo_numeros}`
242 `\begin{tabular}{ccc}` % Define a quantidade de colunas
243 `\hline` % Linha superior
244 `\textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\` % Cabeçalhos
245 `\hline` % Linha média
246 `1 & 2 & 3 \\` % Primeira linha de dados
247 `\hline`

```

248 4 & 5 & 6 \\ % Segunda linha de dados
249 \hline
250 7 & 8 & 9 \\ % Terceira linha de dados
251 \hline
252 10 & 11 & 12 \\ % Quarta linha de dados
253 \hline % Linha inferior
254 \end{tabular}
255 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
256 \end{table}
257
258
259
260 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
261
262 \begin{figure}[ht]
263 \centering % Centraliza a figura
264 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
265 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
266 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
267 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
268 \end{figure}
269
270 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
271
272 \begin{quadro}[ht]
273 \centering
274 \caption{Exemplo de Quadro}
275 \label{qua:exemplo}
276 \begin{tabular}{|l|l|}
277 \hline
278 \textbf{Ilustração} & \textbf{Delimitação} \\
279 \hline
280 Tabela & Linhas apenas no topo e na base. \\
281 \hline
282 Quadro & Linhas em todas as bordas. \\
283 \hline
284 \end{tabular}
285 \fonte{Elabora\c{c}\~ao pr\'opria.}
286 \end{quadro}
287
288
289
290
291
292 \section{Tabelas mais elegantes}
293
294 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
295
296 \begin{table}[ht]
297 \centering % Centraliza a tabela
298 \caption{Exemplo de Tabela de Números mais elegantes}
299 \label{tab:exemplo_numerosbom}
300 \begin{tabular}{ccc} % Define a quantidade de colunas
301 \toprule % Linha superior

```

```

302 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
303 \midrule % Linha média
304 1 & 2 & 3 \\ % Primeira linha de dados
305 4 & 5 & 6 \\ % Segunda linha de dados
306 7 & 8 & 9 \\ % Terceira linha de dados
307 10 & 11 & 12 \\ % Quarta linha de dados
308 \bottomrule % Linha inferior
309 \end{tabular}
310 \fonte{Elabora\c{c}\~ao pr\'opria.}
311 \end{table}
312
313 \section{Tabelas Longas ou Largas}
314
315 Se sua tabela é muito longa ou larga, existem várias opções.
316 \begin{itemize}
317     \item alterar o tamanho da letra
318     \item Usar o longtable
319     \item rodar a tabela, fazendo ela em \textit{landscape}
320     \item fazer a tabela dentro de um minibox
321 \end{itemize}
322
323
324 \subsection{Tabelas largas demais}
325
326 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
327
328 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
329
330 \begin{table}[ht]
331 \centering % Centraliza a tabela
332 \caption{Exemplo de Tabela Larga com Fonte Menor}
333 \label{tab:tabela_largafns}
334 \footnotesize % Aplica uma fonte menor para a tabela
335 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
336 \toprule
337 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} & \textbf{Col. 4} & \textbf{Col. 5} & \textbf{Col. 6} & \textbf{Col. 7} \\
338 \midrule
339 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
340 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
341 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
342 \bottomrule
343 \end{tabular}
344 \fonte{Elabora\c{c}\~ao pr\'opria.}
345 \end{table}
346
347 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
348
349 \begin{table}[ht]
350 \centering
351 \caption{Exemplo de Tabela Redimensionada}
352 \label{tab:examplerb}
353 \resizebox{\textwidth}{!}{%
354 \begin{tabular}{llll}
355 \toprule

```

```

356 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
357 \midrule
358 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
359 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
360 \bottomrule
361 \end{tabular}%
362 }
363 \fonte{Elabora\c{c}\~ao pr\'opria.}
364 \end{table}
365
366
367 Para rodar uma tabela muito larga em 90 graus no LaTeX, voc\ea pode usar o pacote \verb*|rotat
368
369 Aqui est\aa um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
370
371 \begin{sidewaystable}
372 \centering
373 \caption{Sua Legenda Aqui}
374 \label{tab:sua_tabela}
375 \begin{tabular}{llll}
376 \toprule
377 Coluna 1 & Coluna 2 & Coluna 3 & \\
378 \midrule
379 Item 1 & Item 2 & Item 3 & \\
380 Item 4 & Item 5 & Item 6 & \\
381 \bottomrule
382 \end{tabular}
383 \fonte{Elabora\c{c}\~ao pr\'opria.}
384 \end{sidewaystable}
385
386 Se a tabela for muito longa, o ambiente \verb|longtable| \xc3\ea o ideal. Ele fornece comandos par
387
388 % Exemplo de tabela longa que se estende por v\arias p\aginas
389 \begin{longtable}{|c|c|c|}
390 % primeiro cabe\c3alho (\xc3\ea o caption)
391 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
392 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
393 \endfirsthead
394 % cabe\c3alho normal
395 \multicolumn{3}{c}%
396 {\table\thetable} -- continua\c3o da p\agina anterior} \\
397 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
398 \endhead
399 % p\ea normal
400 \hline \multicolumn{3}{r|}{Continua na pr\oxima p\agina} \\ \hline
401 \endfoot
402 \hline
403 % \xc3ltimo p\ea
404 \multicolumn{3}{r|}{Continua na pr\oxima p\agina}} \\
405 \hline \hline
406 \endlastfoot
407
408 % Conte\c3do da tabela
409 1 & 2 & 3 \\

```

410 4 & 5 & 6 \\
 411 1 & 2 & 3 \\
 412 4 & 5 & 6 \\
 413 1 & 2 & 3 \\
 414 4 & 5 & 6 \\
 415 1 & 2 & 3 \\
 416 4 & 5 & 6 \\
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 418 4 & 5 & 6 \\
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 443 4 & 5 & 6 \\
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511 4 & 5 & 6 \\
512 1 & 2 & 3 \\
513 4 & 5 & 6 \\
514 1 & 2 & 3 \\
515 4 & 5 & 6 \\
516
517 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas

```

```

518 \end{longtable}
519 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
520
521 \section{Subfiguras (subcaption)}
522
523 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
524
525 \begin{figure}[ht]
526   \centering
527   \begin{subfigure}[b]{0.42\textwidth}
528     \centering\includegraphics[width=\linewidth]{example-image-a}
529     \caption{Primeira vista}\label{fig:sub-a}
530   \end{subfigure}
531   \hfill
532   \begin{subfigure}[b]{0.42\textwidth}
533     \centering\includegraphics[width=\linewidth]{example-image-b}
534     \caption{Segunda vista}\label{fig:sub-b}
535   \end{subfigure}
536   \caption{Figura com duas subfiguras}
537   \label{fig:subs}
538   \fonte{Elabora\c{c}\~ao pr\'opria.}
539 \end{figure}
540
541 Referencie a figura inteira com \verb|\autoref{fig:subs}|. Cada parte tem
542 rótulo próprio: \verb|\autoref{fig:sub-a}| leva à \verb|\autoref{fig:sub-a}|.
543
544 \section{Imagens (graphicx)}
545
546 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
547 \begin{itemize}
548   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
549   \item \verb|[angle=90]| --- rotação;
550   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direi}
551 \end{itemize}
552 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{t}
553
554 \section{Introdução ao tikz}
555
556 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
557
558 \begin{figure}[ht]
559   \centering
560   \begin{tikzpicture}
561     \node[draw, rounded corners] (a) at (0,0) {Início};
562     \node[draw, rounded corners] (b) at (4,0) {Fim};
563     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
564   \end{tikzpicture}
565   \caption{Diagrama simples em tikz}
566   \label{fig:tikz}
567   \fonte{Elabora\c{c}\~ao pr\'opria.}
568 \end{figure}
569
570 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|ti
571

```

```

572 \section{Float ou direto no texto?}
573
574 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \L
575
576 \chapter{Using BibLaTeX}
577 \label{cap:biblatex}
578
579 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
580
581 \textbf{É} obrigatório compilar com \texttt{biber} (e não com \hologo{BibTeX}). A sequênci
582
583 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
584
585 \begin{table}[h]
586 \caption{Exemplos de cita\c c\~oes utilizando o comando padr\~ao
587 \texttt{\textbackslash cite} do \LaTeX\ e
588 o comando \texttt{\textbackslash citep},
589 disponível pela opção \texttt{natbib} do BibLaTeX.}
590 \label{tab:citation}
591 \centering
592 {\footnotesize
593 \begin{tabular}{|c|c|c|}
594 \hline
595 Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
596 \hline
597 Livro & \cite{book-example} & \citep{book-example}\\
598 Artigo & \cite{article-example} & \citep{article-example}\\
599 Relatório & \cite{techreport-example} & \citep{techreport-example}\\
600 Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
601 Anais de Congresso & \cite{inproceedings-example} &
602 \citep{inproceedings-example}\\
603 Séries & \cite{incollection-example} & \citep{incollection-example}\\
604 Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
605 Dissertação de mestrado & \cite{mastersthesis-example} &
606 \citep{mastersthesis-example}\\
607 Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
608 \hline
609 \end{tabular}}
610 \fonte{Elabora\c c\~ao pr\'opria.}
611 \end{table}
612
613 \begin{table}[h]
614 \caption{Exemplos de cita\c c\~oes utilizando o comando padr\~ao
615 \texttt{\textbackslash cite} do \LaTeX\ e
616 o comando \texttt{\textbackslash citet},
617 disponível pela opção \texttt{natbib} do BibLaTeX. Além disso, usando o booktabs.}
618 \label{tab:citation1}
619 \centering
620 {\footnotesize
621 \begin{tabular}{ccc}
622 \toprule
623 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
624 \midrule
625 Livro & \cite{book-example} & \citet{book-example}\\

```

```

626 Artigo & \cite{article-example} & \citet{article-example}\\
627 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
628 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
629 Anais de Congresso & \cite{inproceedings-example} &
630 \citet{inproceedings-example}\\
631 Séries & \cite{incollection-example} & \citet{incollection-example}\\
632 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
633 Dissertação de mestrado & \cite{mastersthesis-example} &
634 \citet{mastersthesis-example}\\
635 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
636 \bottomrule
637 \end{tabular}}
638 \fonte{Elabora\c{c}\~ao pr'opria.}
639 \end{table}
640
641 \section{Comandos de citação}
642
643 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
644
645 \subsection{Parâmetro opcional: página e “tradução nossa”}
646
647 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
648
649 \subsection{Citação de citação (\emph{apud})}
650
651 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
652
653 \subsection{Múltiplas bibliografias}
654
655 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
656
657 \section{Tipos de entrada bibliográfica}
658
659 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
660
661 \subsection{Tipos acadêmicos principais}
662 \begin{itemize}
663 \item Livro / monografia --- \verb|@book| (\verb|@livro|)
664 \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \verb|@journal|
665 \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
666 \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
667 \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
668 \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periód
669 \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
670 \end{itemize}
671
672 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
673 \begin{itemize}
674 \item Patente --- \verb|@patent| (\verb|@patente|)
675 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato|
676 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\verb|@partitura|)
677 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
678 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb|@filme|)
679 \end{itemize}

```

```

680
681 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
682 \begin{itemize}
683   \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j|
684   \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadstv|, \verb|@serietv|)
685 \end{itemize}
686
687 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
688
689 \chapter{Alguns outros exemplo úteis}
690
691 \begin{tcolorbox}[title=Meu Textbox]
692 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
693 \end{tcolorbox}
694
695 \begin{tcolorbox}
696 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
697 \end{tcolorbox}
698
699 \begin{figure}[ht]
700   \centering
701   \begin{tikzpicture}
702     \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{}}
703     \begin{scope}[x={(image.south east)},y={(image.north west)}]
704       % Definindo o textbox dentro da figura
705       \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=
706         \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
707           Este textbox fala sobre como inserir um textbox dentro de uma figura usando
708         \end{tcolorbox}
709       ];
710     \end{scope}
711   \end{tikzpicture}
712   \caption{Figura com Textbox}
713   \label{fig:figura_com_textbox1}
714   \fonte{Elabora\c{c}\~ao pr\'opria.}
715 \end{figure}
716
717
718 \begin{figure}[ht]
719   \centering
720   \begin{tcolorbox}
721 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
722 \end{tcolorbox}
723   \caption{Figura com Textbox simples}
724   \label{fig:figura_com_textbox}
725   \fonte{Elabora\c{c}\~ao pr\'opria.}
726 \end{figure}
727
728 \section{Listagens de código e algoritmos}
729
730 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
731
732 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
733 def saudacao(nome):

```

```

734     # imprime uma saudação acentuada
735     print(f"Olá, {nome}!")
736
737 saudacao("COPPE")
738 \end{python}
739 \source{Elaboração própria.}
740
741 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
742
743 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
744 Olá, COPPE!
745 \end{Verbatim}
746
747 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|.
748 Há também \verb|\pythoninline| e \verb|\javainline| para código no meio do
749 texto, e o ambiente \verb|Verbatim| (com as opções acima) para a saída de programas. Para al
750
751 \begin{algorithm}[H]
752 \caption{Soma dos elementos de um vetor}
753 \label{alg:soma}
754 \Dados{vetor $v$ com $n$ elementos}
755 \Resultado{soma $$ dos elementos de $v$}
756 $$ \leftarrow 0;
757 \Para{$i$ \leftarrow 1$ \Ate{$n$}}{
758     $$ \leftarrow s + v[i];
759 }
760 \Retorna $$;
761 \end{algorithm}
762
763 \chapter{Escrevendo matemática}
764
765 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr
766
767 \section{Alinhamento (align)}
768 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
769 \begin{align}
770     f(x) &= (x+1)^2 \\
771         &= x^2 + 2x + 1 .
772 \end{align}
773
774 \section{Matrizes (matrix)}
775 O \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
776 \begin{equation}
777     A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
778     \quad
779     \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
780 \end{equation}
781
782 \section{Subequações (subequations)}
783 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
784 \begin{subequations}
785 \begin{align}
786     a &= b + c , \\
787     d &= e - f .

```

```

788 \end{align}
789 \end{subequations}
790
791 \section{Estruturas alinhadas (array)}
792 0 \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
793 \begin{equation}
794 |x| = \left\{ \begin{array}{l}
795 x, \quad \text{se } x \geq 0, \\
796 -x, \quad \text{caso contrário.}
797 \end{array} \right.
798 \end{equation}
799
800 \section{Operadores (\textbackslash DeclareMathOperator)}
801 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
802 \begin{verbatim}
803 \usepackage{amsmath}
804 \DeclareMathOperator{\RMSE}{RMSE}
805 \DeclareMathOperator{\diag}{diag}
806 \end{verbatim}
807 Assim:
808 \begin{itemize}
809 \item \verb|\$RMSE(y,\hat{y})$| produz  $\text{RMSE}(y, \hat{y})$ ;
810 \item \verb|\$diag(\lambda_1, \lambda_2)$| produz
811  $\text{diag}(\lambda_1, \lambda_2)$ .
812 \end{itemize}
813
814 \section{Unicode e LuaLaTeX (unicode-math)}
815 \label{sec:unicode-math}
816 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
817 \begin{verbatim}
818 \usepackage{amsmath} % a classe já carrega
819 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
820 \setmathfont{Latin Modern Math} % ou STIX Two Math
821 \end{verbatim}
822 0 código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
823
824 \chapter{Referências cruzadas, notas e índices}
825 \label{cap:refs}
826
827 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
828
829 \section{Rótulos e referências cruzadas}
830
831 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
832
833 \section{Notas de rodapé}
834
835 Use \verb|\footnote{texto}| \footnote{Como esta nota, que a classe imprime em espaçamento s
836
837 \section{hyperref e marcadores}
838
839 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref|
840
841 \section{URLs}

```

```

842
843 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
844 use \verb|\href{https://exemplo.org}{texto}|. Evite digitar URLs como texto comum --- elas n
845
846 \section{Índice remissivo}
847
848 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
849
850 \section{Traduções (NBR 10520)}
851
852 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
853
854 \chapter{Truques de \LaTeX}
855
856 \section{Comentários mágicos (\texttt{\% !TeX})}
857 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
858 \begin{verbatim}
859 % !TeX program = lualatex
860 % !TeX encoding = UTF-8
861 % !TeX spellcheck = pt_BR
862 % !BIB program = biber
863 \end{verbatim}
864 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
865
866 \section{Espaço engolido}
867 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
868
869 \section{Escopo (grupos)}
870 Chaves \verb|\{...\}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
871
872 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash m
873 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
874
875 \section{\texttt{latexmkrc}, perl e Overleaf}
876 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
877
878 \chapter{Método Proposto}
879 \chapter{Resultados e Discuss\~oes}
880
881 \section{Algumas Demonstra\c c\~oes}
882
883 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha
884 \symbl[beta]{Beta}{A palavra Beta more e corrigida}
885 \symbl[zbeta]{\beta}{A letra $\beta$ corrigida}
886 \symbl[beta]{A palavra beta}
887 \symbl[alpha]{Alpha}{A palavra Alpha}
888 \symbl[zalpha]{\alpha}{A letra $\alpha$ corrigida}
889 \symbl[marco]{Marco}{A palavra Marco corrigida}
890
891
892 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exempl
893 \abbrev{GoT}{Game of Thrones}
894 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
895 \abbrev[iot]{IoT}{IoT ordenado como iot}

```



```

896 \abbrev[IOT]{IoT}{IoT ordenado como IoT}
897 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
898 \abbrev{IoT}{IoT com ordenação default}
899 \abbrev[ITU]{ITU}{ITU mesmo}
900
901 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
902 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
903
904 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto
905
906
907
908 \chapter{Conclusões}
909
910
911 \backmatter
912
913 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
914 % partitura --- declarados via sinônimos em português em example.bib.
915 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
916 \nocite{jurisprudencia-example}
917 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
918 \nocite{jornalcaderno-example,jornal-example}
919 \printbibliography
920
921
922
923 \appendix
924
925 \chapter{Um apêndice}
926
927 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
928
929 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
930
931
932 \chapter{Outro apêndice}
933
934 \annex
935
936
937 \chapter{Um Anexo}
938 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
939
940
941
942 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
943
944 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
945
946
947 \chapter{Outro Anexo}
948
949

```

```

950 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
951
952 \end{document}
953
954 </example>

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

955 <*class>
956 \def\filename{coppe.dtx}
957 \def\fileversion{v4.0}
958 \def\filedate{2026/05/28}
959 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
960 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
961 % Save the class version and date into private macros immediately, before any
962 % required package can redefine the very common \fileversion / \filedate
963 % global names (LaTeX has no convention reserving them for the host class).
964 % \coppe@version / \coppe@date are what user-facing macros (e.g.
965 % \coppetexfinalpage) should reference.
966 \let\coppe@version\fileversion
967 \let\coppe@date\filedate
968 % Document date stored in private slots so \date can set them without ever
969 % touching TeX's \month/\year primitives (which would break \today). They
970 % default to the current system month/year, so a document without \date
971 % still renders sensibly on the cover and folha de rosto.
972 \edef\coppe@docmonth{\the\month}
973 \edef\coppe@docyear{\the\year}
974 \LoadClass[12pt,a4paper,oneside]{book}
975 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
976 % the END of the class (after every class hook is defined), in that order:
977 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
978 % it patches (TOC, refs, lists) are already defined, and biblatex must come
979 % after hyperref so its citation links and back-references work.
980 \RequirePackage{hyphenat}
981 \RequirePackage{lastpage}
982 \RequirePackage{ifthen}
983 \RequirePackage{graphicx}
984 \RequirePackage{tabularx}
985 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
986 \RequirePackage{longtable} % tabelas que quebram entre páginas
987 \RequirePackage{etoolbox}
988 \RequirePackage{ltxcmds}
989 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
990 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
991 \def\TeX{\hologo{TeX}}
992 \def\LaTeX{\hologo{LaTeX}}
993 \def\LaTeXe{\hologo{LaTeXe}}
994 \def\BibTeX{\hologo{BibTeX}}
995 \def\pdfLaTeX{\hologo{pdfLaTeX}}
996 \def\pdfTeX{\hologo{pdfTeX}}
997 \def\LuaLaTeX{\hologo{LuaLaTeX}}
998 \def\XeLaTeX{\hologo{XeLaTeX}}

```

```

999 \def\MiKTeX{\hologo{MiKTeX}}
1000 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
1001 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
1002 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
1003 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
1004 \RequirePackage{iftex}
1005 \ifPDFTeX
1006   \RequirePackage[utf8]{inputenc}
1007   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
1008     Although slower, you should try compiling with LuaLaTeX}}
1009 \fi
1010 % lmodern MUST come with the T1 encoding. Without it, T1 falls back to the EC
1011 % fonts, which exist only as METAFONT bitmaps in most installations, and the
1012 % whole document is typeset in Type3 bitmap fonts. Measured before this change:
1013 % example.pdf carried 87 Type3 fonts against 6 Type1, and coppe.pdf 133 against
1014 % 2. The consequences are not cosmetic:
1015 %
1016 % * the text renders blurred at any zoom and prints poorly;
1017 % * copying and searching text in the PDF is unreliable;
1018 % * PDF/A -- required for deposit by 2.2(d) of the 2026 manual -- is not
1019 %   attainable with bitmap text.
1020 %
1021 % Latin Modern is the vector successor of Computer Modern, metrically and
1022 % visually near-identical, and ships with every current distribution, so the
1023 % page layout does not move.
1024 \RequirePackage{lmodern}
1025 \RequirePackage[T1]{fontenc}
1026 % Math support. amsmath is universal and works on every engine, so the class
1027 % brings it in unconditionally (it must precede hyperref, which is loaded at
1028 % the end of the class). amssymb is the legacy AMS symbol package, built
1029 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
1030 % XeLaTeX) it clashes with unicode-math, which already provides all of its
1031 % symbols through Unicode math codes. So amssymb is loaded ONLY under
1032 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
1033 % unicode-math (the class does not auto-load unicode-math because choosing
1034 % an OpenType math font is a typographic decision that belongs to the author).
1035 \RequirePackage{amsmath}
1036 \ifPDFTeX
1037   \RequirePackage{amssymb}
1038 \else
1039   % Unicode engine: defer the check to end of document. Only warn if the
1040   % user did NOT add either unicode-math (the modern OpenType-math route)
1041   % or amssymb (the legacy route, which they would have loaded manually).
1042   % This avoids nagging authors who already followed the documented
1043   % \usepackage{unicode-math} workflow.
1044   \AtEndDocument{%
1045     \@ifpackageloaded{unicode-math}{}{%
1046       \@ifpackageloaded{amssymb}{}{%
1047         \ClassWarningNoLine{coppe}{%
1048           You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1049           neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1050           (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1051           Add to your preamble:\MessageBreak
1052           \space\space\string\usepackage{unicode-math}\MessageBreak

```

```

1053         \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1054         Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1055         with the OpenType math font setup}}}}
1056 \fi
1057 % Margins per the SiBI manual, 9th rev. ed. (2026), 2.3: left 3cm, top 3cm,
1058 % right 2cm, bottom 2cm -- ONE-SIDED. The 2025 edition also specified verso
1059 % margins (right 3cm / left 2cm) and recommended printing the body on both
1060 % sides; the 2026 revision DELETED the verso margins entirely, following CEPG
1061 % Resolution 246/2023, which ended the printed deposit. A digital-only document
1062 % has no gutter, so the mirrored layout (previously inner=outer=2cm plus a 1cm
1063 % bindingoffset) no longer has any normative basis and is now opt-in via the
1064 % \texttt{twoside} class option, handled after \ProcessOptions below.
1065 %
1066 % headheight=15pt is given here (not patched later) so geometry's own page
1067 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1068 % height to host the running head, and the standard 12pt default triggers
1069 % its "headheight is too small" warning. Letting geometry know up front
1070 % keeps \topmargin a derived value -- we never touch it by hand.
1071 % (geometry has no \texttt{oneside} key -- only \texttt{twoside}; with the key
1072 % absent it follows the class, which \LoadClass set to one-sided above.)
1073 % headsep is given explicitly to place the folio. 2.7 measures it the way it
1074 % measures the right margin -- to the digits themselves: "a 2cm da borda
1075 % superior, ficando o ultimo algarismo a 2cm da borda direita". The right edge
1076 % already landed at 2.02cm; the top of the digits sat at 1.87cm, too high.
1077 % geometry derives \topmargin from top - 1in - headheight - headsep, so
1078 % shortening headsep lowers the header and leaves the body top untouched:
1079 % topmargin + headheight + headsep is 13.088pt either way, and 3cm is where the
1080 % body still starts. The value is CALIBRATED, not computed -- the folio sits a
1081 % little lower than the arithmetic alone predicts, because fancyhdr's box does
1082 % not put the digits flush with the top of \headheight. Two measured points,
1083 % taken from the rendered page rather than from the PDF's font metrics:
1084 %   headsep 18.07pt -> digits at 1.8669cm   headsep 14.29pt -> 2.0616cm
1085 % which put 2.0000cm at 15.49pt while the folio was still set at the body size.
1086 % It is \footnotesize now (2.2(b) puts pagination in the smaller font), the
1087 % digits are shorter, and their top therefore sits lower for the same baseline:
1088 % re-measured, 15.49pt gave 2.0659cm and the value moved to 16.87pt. That is
1089 % precisely why this is a measured constant and not a computed one -- change
1090 % the folio's size or headheight and it has to be measured again, not scaled.
1091 % Verified on the rendered page: the top of the digits and their right edge come
1092 % out at the same 2.019cm, and the right margin is 2cm by construction, so that
1093 % shared 0.019cm is the ruler's own bias and not a residual error.
1094 \RequirePackage[a4paper,headheight=15pt,headsep=16.87pt,%
1095 top=3cm,bottom=2cm,left=3cm,right=2cm]{geometry}
1096 % The first paragraph of every section must be indented like any other. LaTeX's
1097 % default is the Anglo-Saxon convention -- no indent after a heading -- which is
1098 % not what Brazilian typography (nor the SiBI manual's own examples) uses.
1099 % indentfirst was mentioned only as a commented suggestion in the example
1100 % preamble, so every document built with this class came out with the first
1101 % paragraph of each section flush left. Loading it in the class fixes it for
1102 % everybody instead of asking each author to remember.
1103 %
1104 % Loaded BEFORE titlesec so titlesec's \@afterindent handling sees the patched
1105 % \@afterindentfalse.
1106 % Long unbreakable tokens -- \verb|...| in running prose, command names, URLs --

```

```

1107 % cannot be hyphenated, so TeX overfills the line rather than stretching the
1108 % spaces. example.tex alone produced 17 overfull boxes this way, the worst 86pt,
1109 % which is the "text does not always respect the margins" a recent review
1110 % reported. \emergencystretch gives TeX a final pass in which it may stretch
1111 % interword spaces instead of running into the margin. It cannot save a single
1112 % token longer than the measure, but it clears the ordinary cases.
1113 \setlength\emergencystretch{3em}
1114 \RequirePackage[indentfirst]
1115 \RequirePackage{titlesec}
1116 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1117 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
1118 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1119 % keep "N Title". \appendix and \annex raise the flag (see below).
1120 \newif\if@coppeappendix \@coppeappendixfalse
1121 \titleformat{\chapter}[hang]
1122   {\normalfont\Large\bfseries}
1123   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1124   {1ex}\MakeUppercase}
1125 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1126 % 2.6 says the numeric indicative of a section precedes its title flush left,
1127 % and that titles WITHOUT a numeric indicative are centred -- and then lists
1128 % them, "apendice(s)" and "anexo(s)" among them. A letter is not a numeric
1129 % indicative, and the manual sets its own Annex A centred. The chapter format is
1130 % therefore re-issued when \appendix or \annex is called: same label, same
1131 % travessao, but a centred block instead of a flush-left hang. Numbered chapters
1132 % are untouched -- they DO carry a numeric indicative and stay flush left.
1133 \newcommand\coppe@appendixheadings{%
1134   \titleformat{\chapter}[block]
1135     {\normalfont\Large\bfseries\filcenter}
1136     {\MakeUppercase{\appendixname}\ \thechapter\ \textendash}
1137     {1ex}\MakeUppercase}%
1138 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}}
1139 \let\coppe@orig@appendix\appendix
1140 \renewcommand\appendix{%
1141   \coppe@orig@appendix\@coppeappendixtrue\coppe@appendixheadings}
1142 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1143 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1144 % which macro is written (so it is correct at .toc-render time even though the
1145 % appendixname has changed back); the chapter letter is baked in at write time.
1146 \newif\if@coppeannex \@coppeannexfalse
1147 \newcommand\coppe@tocapp[1]{%
1148   \MakeUppercase{\coppestring{appendix}}}%
1149   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1150 \newcommand\coppe@tocanx[1]{%
1151   \MakeUppercase{\coppestring{annex}}}%
1152   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1153 \def\@chapter[#1]#2{%
1154   \ifnum \c@secnumdepth >\m@ne
1155     \if@mainmatter
1156       \refstepcounter{chapter}%
1157       \typeout{\@chapapp\space\thechapter.}%
1158       \if@coppeappendix
1159         \if@coppeannex
1160           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%

```

```

1161         \else
1162         \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1163         \fi
1164     \else
1165         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1166     \fi
1167 \else
1168     \addcontentsline{toc}{chapter}{#1}%
1169 \fi
1170 \else
1171     \addcontentsline{toc}{chapter}{#1}%
1172 \fi
1173 \chaptermark{#1}%
1174 \addtocontents{lof}{\protect\addvspace{10\p@}}%
1175 \addtocontents{lot}{\protect\addvspace{10\p@}}%
1176 \if@twocolumn
1177     \@topnewpage[\@makechapterhead{#2}]%
1178 \else
1179     \@makechapterhead{#2}%
1180     \@afterheading
1181 \fi}
1182 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1183 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1184 \titleformat{name=chapter,numberless}[hang]
1185     {\normalfont\Large\bfseries\filcenter}{\Opt}{\MakeUppercase}
1186 \titlespacing*{name=chapter,numberless}{Opt}{10pt}{1.5\baselineskip}
1187 % section: ALL CAPS, normal weight -- the \bfseries here contradicted this very
1188 % comment until v4.1, and made the secondary level as heavy as the primary one,
1189 % defeating the "destaque gradativo" of 2.6. The gradation is now
1190 % chapter = ALL CAPS bold, section = ALL CAPS regular, subsection = Title Case
1191 % bold, subsubsection = Title Case bold italic, as read from the manual.
1192 \titleformat{\section}
1193     {\normalfont\large}{\thesection}{1ex}{\MakeUppercase}
1194 % subsection: bold (author writes Title Case).
1195 \titleformat{\subsection}
1196     {\normalfont\bfseries}{\thesubsection}{1ex}{\}
1197 % subsubsection: bold italic.
1198 \titleformat{\subsubsection}
1199     {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{\}
1200 % paragraph = quinary section: italic, the author capitalising only the first
1201 % word. 2.6 recommends going as deep as the quinary, and the annotation on the
1202 % 2025 manual (specs/ANOTACOES.md) describes exactly this shape.
1203 %
1204 % book makes \paragraph a RUN-IN heading, so the text continued on the same
1205 % line -- 2.6 requires "o texto deve ser iniciado em outra linha". The positive
1206 % "after" spacing in \titlespacing turns it into a displayed heading.
1207 \titleformat{\paragraph}[hang]
1208     {\normalfont\itshape}{\theparagraph}{1ex}{\}
1209 \titlespacing*{\paragraph}{Opt}{2.5ex plus 1ex minus .2ex}{1.5ex plus .2ex}
1210 % book defaults secnumdepth to 2, so \subsubsection and \paragraph came out
1211 % UNNUMBERED however carefully \titleformat named \thesubsubsection. 2.6 has
1212 % the numbering following the whole chain down to the quinary, and the manual's
1213 % own sumario is four levels deep, so tocdepth follows.
1214 \setcounter{secnumdepth}{4}

```

```

1215 \setcounter{tocdepth}{4}
1216 % 2.6: "no sumario, as secões devem ser grafadas conforme apresentadas no corpo
1217 % do trabalho". The body sets chapter and section in ALL CAPS (via the
1218 % \MakeUppercase in their \titleformat), but that case change belongs to the
1219 % printed heading alone -- the .toc keeps the title as typed, so the sumario
1220 % came out in ordinary case and contradicted the body.
1221 %
1222 % \l@chapter and \l@section are the two TOC line formatters; wrapping their
1223 % entry argument is enough, and leaves \l@subsection and below untouched, which
1224 % is what we want since those levels are NOT uppercased in the body either.
1225 % The entry carries \numberline{N} plus the title; \MakeUppercase passes the
1226 % digits through unchanged and leaves robust commands (including hyperref's
1227 % link wrappers) intact.
1228 \let\coppe@orig\lchapter\lchapter
1229 \renewcommand\lchapter[2]{\coppe@orig\lchapter{\MakeUppercase{#1}}{#2}}
1230 \let\coppe@orig\lsection\lsection
1231 \renewcommand\lsection[2]{\coppe@orig\lsection{\MakeUppercase{#1}}{#2}}
1232 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1233 % "Figura 1 -- título" travessão separator (sources handled by \source). ---
1234 \RequirePackage{caption}
1235 % justification=centering makes MULTI-LINE captions centre like the one-line
1236 % ones. Without it, caption's singlelinecheck centres a caption that fits on one
1237 % line and justifies one that wraps, so in a document with mixed caption lengths
1238 % a couple of tables looked left-aligned while the rest looked centred -- which
1239 % is what a recent review noticed about tables 5.1 and 5.2. 2.10 does not
1240 % prescribe the alignment; it does not tolerate two alignments either.
1241 \captionsetup{font=footnotesize,labelsep=endash,justification=centering}
1242 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1243 % caption so it picks up the ABNT captionsetup above.
1244 \RequirePackage{subcaption}
1245 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1246 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1247 % at the bottom is mandatory). The float package's plaintop style forces the
1248 % caption above the body for figure, table and the new quadro float. -----
1249 \RequirePackage{float}
1250 \floatstyle{plaintop}
1251 \restylefloat{figure}
1252 \restylefloat{table}
1253 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1254 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1255 % unlike a tabela (rules on top and bottom only).
1256 \providecommand{\quadroname}{\coppestring{frame}}
1257 \providecommand{\listquadroname}{\coppestring{listframe}}
1258 \newfloat{quadro}{htbp}{loq}[chapter]
1259 \floatname{quadro}{\quadroname}
1260 \providecommand{\quadroautorefname}{\quadroname}
1261 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1262 % (float's own \l@quadro lays the number/title out incorrectly).
1263 \let\l@quadro\l@figure
1264 \newcommand\listofquadros{%
1265   \chapter*{\listquadroname}%
1266   \coppe@listtoc{\listquadroname}%
1267   \mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1268   \@starttoc{loq}}

```

```

1269 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1270 % float as a smaller, single-spaced, centred line; it does NOT advance any
1271 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1272 % the label is "Fonte" (pt) / "Source" (en).
1273 \newcommand{\cpsourcenam}{\coppestring{source}}
1274 \newcommand{\cpsource}[1]{%
1275   \par\addvspace{\smallskipamount}%
1276   {\footnotesize\singlespacing\centering\cpsourcenam: #1\par}}
1277 \let\cpfonte\cpsource
1278 \@ifundefined{source}{\let\source\cpsource}{}%
1279 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1280 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1281 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1282 \newcommand{\newcoppefloat}[3]{%
1283   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1284   \floatname{#1}{#2}%
1285   \expandafter\newcommand\csname listof#1s\endcsname{%
1286     \chapter*{#3}\coppe@listtoc{#3}%
1287     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1288     \@starttoc{lo#1}}}%
1289 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1290 % (filete) from the left margin; footnotesize is the book default. -----
1291 \let\coppe@orig@footnotetext\@footnotetext
1292 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1293 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1294 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1295 % (manual2024 2.6 / proposta P24). -----
1296 \RequirePackage{enumitem}
1297 \newlist{alineas}{enumerate}{2}
1298 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1299 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1300 % --- Page numbering: arabic in the top OUTER corner on textual pages
1301 % (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1302 % \frontmatter (unnumbered by default; roman with the numeraisromanos
1303 % class option). -----
1304 % Every list goes through \@starttoc, which allocates a stream the first time
1305 % it sees an extension. That is where the ceiling is hit in practice, so that
1306 % is where the class checks -- one \ifnum before the allocation, and an error
1307 % that names the three ways out instead of the kernel's bare "No room".
1308 %
1309 % \count17 is the write allocation counter, and 15 is the last usable stream.
1310 % Under LuaTeX there are 128 and the counter is managed differently, so the
1311 % check is skipped there; the same is true when morewrites is in charge.
1312 % The wrap waits for \begin{document} on purpose: hyperref redefines
1313 % \@starttoc as well, and it is loaded at the very end of this class, so a
1314 % \let taken here would be thrown away.
1315 \newcommand\coppe@checkwrites[1]{%
1316   \ifdefined\directlua\else
1317     \if@coppemorewrites\else
1318       \@ifundefined{tf#1}{%
1319         \ifnum\count17>14
1320           \ClassError{coppe}{Acabaram os 16 fluxos de escrita do TeX}{%
1321             Este trabalho pede mais listas do que o pdfTeX consegue manter
1322             abertas ao mesmo tempo.\MessageBreak

```



```

1323      Tres saidas, nesta ordem de preferencia:\MessageBreak
1324      (1) compile com lualatex, que tem 128 fluxos;\MessageBreak
1325      (2) passe a opcao 'morewrites' a classe coppe;\MessageBreak
1326      (3) remova uma lista que o trabalho nao usa.\MessageBreak
1327      A lista que faltou agora foi a de extensao '#1', mas o problema
1328      nao e dela: e a soma de todas.}%
1329      \fi}{}%
1330      \fi
1331      \fi}
1332 % The wrapping itself lives in a macro of its own rather than inside the hook:
1333 % \AtBeginDocument stores its argument by expansion, and a parameter written
1334 % there does not survive it -- the document died with "Parameters must be
1335 % numbered consecutively" at \begin{document}. One level of nesting, one
1336 % doubling, and the hook only has to call it.
1337 \newcommand\coppe@wrapstarttoc{%
1338   \let\coppe@orig@starttoc@starttoc
1339   \def\@starttoc##1{\coppe@checkwrites{##1}\coppe@orig@starttoc{##1}}
1340 \AtBeginDocument{\coppe@wrapstarttoc}
1341 \RequirePackage{fancyhdr}
1342 % \headheight is set to 15pt by geometry up above (see the geometry call),
1343 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1344 % 2.2(b) lists what is set in "fonte menor e uniforme": long quotations,
1345 % footnotes, PAGINATION, and the captions of illustrations and tables. The
1346 % folio was the one item on that list still coming out at the body size of 12pt.
1347 % It is \footnotesize now, which is the same smaller size the other three
1348 % already use -- "uniforme" is part of the requirement, not just "menor".
1349 % Pre-textual lists reach the sumario only under the listasnosumario option
1350 % (see the option's declaration for the reason). Post-textual material calls
1351 % \addcontentsline directly and is unaffected.
1352 \newcommand\coppe@listtoc[1]{%
1353   \if@coppe@listastoc\addcontentsline{toc}{chapter}{#1}\fi
1354 % \coppe@glosspre marks the two PRE-textual uses of the glossary environment --
1355 % the list of abbreviations and the list of symbols -- apart from the genuine
1356 % post-textual Glossario, which shares the same environment but must appear in
1357 % the sumario.
1358 \newif\if@coppeglosspre \@coppeglossprefalse
1359 \newcommand\coppe@folio{\footnotesize\thepage}
1360 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}%
1361   \renewcommand{\headrulewidth}{0pt}}
1362 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1363 % number may leak onto them (e.g. the verso right after the cover).
1364 \renewcommand{\cleardoublepage}{%
1365   \clearpage
1366   \if@twoside
1367     \ifodd\c@page\else
1368       \hbox{}\thispagestyle{empty}\newpage
1369     \if@twocolumn\hbox{}\newpage\fi
1370   \fi
1371   \fi}
1372 \def\CoppeTeX{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1373 p\kern-.025em e}\kern-.08em
1374 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
1375 \newboolean{maledoc}
1376 \setboolean{maledoc}{false}

```

```

1377 %
1378 % Class options.
1379 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1380 % fixed language slots:
1381 % * main -- the language of the body, captions, and the main abstract.
1382 %          Set by a class option: brazilian (default), english,
1383 %          spanish, french, italian, or any user-supplied language
1384 %          that ships a coppe-lang-<name>.def file.
1385 % * foreign -- the language used by the foreignabstract environment and
1386 %          by \foreign@... registers. Defaults to english (or to
1387 %          brazilian when main=english) for back-compat.
1388 % * third -- optional, loaded on demand with \usecoppelanguage{<name>}
1389 %          in the preamble (for citations or quotations in a third
1390 %          language). Has no class option of its own.
1391 % Note: the legacy \if@english boolean (defined and toggled here in
1392 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1393 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.ltx)
1394 % tests it. Code that needs to branch on the main language should test
1395 % \ifx\coppe@mainlang\coppe@englishstr instead.
1396 \def\coppe@mainlang{brazilian}
1397 \def\coppe@foreignlang{english}
1398 \DeclareOption{brazilian}{%
1399   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1400 \DeclareOption{english}{%
1401   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1402 \DeclareOption{spanish}{%
1403   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1404 \DeclareOption{french}{%
1405   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1406 \DeclareOption{italian}{%
1407   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1408 \DeclareOption{msc}{%
1409   \newcommand{\@degree}{M.Sc.}
1410   \newcommand{\@degreename}{Mestrado}
1411   \newcommand{\local@degname}{Mestre}
1412   \newcommand{\foreign@degname}{Master}
1413   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1414   \newcommand\foreign@doctype{Dissertation}
1415 }
1416 \DeclareOption{dsccexam}{%
1417   \newcommand{\@degree}{D.Sc.}
1418   \newcommand{\@degreename}{Doutorado}
1419   \newcommand{\local@degname}{Doutor}
1420   \newcommand{\foreign@degname}{Doctor}
1421   \setboolean{maledoc}{true}
1422   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1423   \newcommand\foreign@doctype{Qualifying Exam}
1424 }
1425 \DeclareOption{mscexam}{%
1426   \newcommand{\@degree}{M.Sc.}
1427   \newcommand{\@degreename}{Mestrado}
1428   \newcommand{\local@degname}{Mestre}
1429   \newcommand{\foreign@degname}{Master}
1430   \setboolean{maledoc}{true}

```

```

1431 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1432 \newcommand\foreign@doctype{Qualifying Exam}
1433 }
1434 \DeclareOption{dsc}{%
1435 \newcommand{\@degree}{D.Sc.}
1436 \newcommand{\@degreename}{Doutorado}
1437 \newcommand{\local@degname}{Doutor}
1438 \newcommand{\foreign@degname}{Doctor}
1439 \newcommand\local@doctype{Tese}
1440 \newcommand\foreign@doctype{Thesis}
1441 }
1442 \newif\if@coppenumeric\@coppenumericfalse
1443 \DeclareOption{numbers}{\@coppenumerictrue}
1444 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1445 % pre-textual pages are counted but not numbered (manual2024 2.7).
1446 \newif\if@copperoman\@copperomanfalse
1447 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the `doublespacing` option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1448 \newif\if@coppedoublespace\@coppedoublespacefalse
1449 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1450 % Two-sided layout is opt-in since v4.1: the 2026 manual dropped the verso
1451 % margins and the deposit is digital (CEPG 246/2023). Given \texttt{twoside},
1452 % the mirrored 3cm inner / 2cm outer layout is restored after \ProcessOptions.
1453 % With no \fichacatalografica file given, the folha adicional leaves a framed
1454 % blank where the ficha goes. The rascunhoficha option fills that blank with the
1455 % class's own LaTeX-composed ficha instead -- useful while writing, never valid
1456 % for deposit: the official ficha comes from the SiBI generator.
1457 % Annex D of the manual draws one signature rule per board member, with the
1458 % identification underneath. The COPPE folha de aprovacao has listed the names
1459 % compactly, without rules, for years. The compact list stays the default and
1460 % the rules are opt-in, so existing theses do not change shape overnight.
1461 % PDF/A output. Opt-in for now: 2.2(d) of the 2026 manual requires the deposited
1462 % file to be PDF/A, but the constraints (embedded fonts, colour profile, no
1463 % encryption, restricted transparency) can trip up a document that pulls in
1464 % arbitrary images or packages, and breaking existing theses to enforce it would
1465 % be worse than letting the author switch it on when depositing.
1466 \newif\if@coppepdfa \@coppepdfafalse
1467 \DeclareOption{pdfa}{\@coppepdfatrue}
1468 \newif\if@coppeassin \@coppeassinfalse
1469 \DeclareOption{assinaturas}{\@coppeassintrue}
1470 % As paginas de resumo listam os ORIENTADORES abaixo do titulo. O coorientador
1471 % nao aparece ali, e isso e tradicao da COPPE, nao exigencia do manual: a
1472 % 3.1.2.1.4 descreve o resumo e nao lista quem assina a orientacao. Como ha
1473 % Programas que preferem ve-lo, a opcao acrescenta os coorientadores nas tres
1474 % paginas de resumo, no mesmo formato da folha de rosto. Sem a opcao nada muda.
1475 \newif\if@coppecoadvresumo \@coppecoadvresumofalse
1476 \DeclareOption{coorientador}{\@coppecoadvresumotrue}
1477 % 3.1.2.1.6 says the sumario enumerates the divisions "na mesma ordem e grafia
1478 % em que a materia nele se sucede" and adds "inclusive os elementos
1479 % pos-textuais" -- and then settles the question by pointing at its own sumario

```

```

1480 % as the model. That sumario opens at "1 INTRODUCAO": the lists of figures, of
1481 % tables, of abbreviations and of symbols come BEFORE the sumario and are not
1482 % listed in it, which is also the rule of NBR 6027 for pre-textual elements.
1483 % The class was adding all of them, so every thesis carried six entries the
1484 % manual's own example does not have. They come out. Post-textual elements --
1485 % Referencias, Glossario, Apendices, Anexos, Indice -- stay, as required.
1486 %
1487 % Programas that want the old behaviour ask for it by name.
1488 \newif\if@coppelistastoc \@coppelistastocfalse
1489 \DeclareOption{listasnosumario}{\@coppelistastoctrue}
1490 \newif\if@copperascunho \@copperascunhofalse
1491 \DeclareOption{rascunhoficha}{\@copperascunhotrue}
1492 \newif\if@coppetwoside \@coppetwosidefalse
1493 \DeclareOption{twoside}{\@coppetwosidetrue}
1494 \DeclareOption{oneside}{\@coppetwosidefalse}
1495 % --- Fluxos de escrita -----
1496 % TeX has SIXTEEN \write streams and no more. Every list costs one for the
1497 % whole run: the .toc, the .lof, the .lot, and one for each list this class
1498 % adds (.loq for quadros, .lol for programas, .loa for algoritmos), plus the
1499 % .abx and .syx behind the lists of abbreviations and of symbols, the .idx of
1500 % the index, hyperref's .out, biblalex's .bcf and the kernel's .aux and friends.
1501 % A work that asks for all of them sits exactly at the ceiling.
1502 %
1503 % The kernel's failure in that situation -- "No room for a new \write" -- names
1504 % neither the cause nor the way out, and lands on whichever list happens to be
1505 % requested last, which is usually \tableofcontents and has nothing to do with
1506 % it. The class therefore checks before the allocation and says what is
1507 % happening (see \coppe@starttoc below).
1508 %
1509 % There are three ways out and the class chooses none of them for the author:
1510 %
1511 % * compile with LuaLaTeX. It has 128 streams; nothing else has to change,
1512 % and the class supports it.
1513 % * let the class load 'morewrites', which multiplexes the streams so the
1514 % ceiling disappears under pdfLaTeX.
1515 % * drop a list the work does not use. Each one returns a stream.
1516 %
1517 % The DEFAULT is the second one, and it is on. An error telling the author to
1518 % pass an option is advice for someone who already knows: a student two days
1519 % from the deposit reads "No room for a new \write" and concludes the class is
1520 % broken. So under pdfTeX the class loads morewrites by itself.
1521 %
1522 % Three things keep that from becoming a burden:
1523 %
1524 % * it is loaded ONLY IF INSTALLED. Missing, the class carries on without it
1525 % and warns; morewrites does not become a hard dependency of a work that
1526 % would never have reached the ceiling anyway.
1527 % * 'semmorewrites' turns it off, for whoever hits a conflict. morewrites
1528 % patches \newwrite, \openout and \write and buffers what is written until
1529 % shipout, so a package that writes a file and reads it back in the SAME run
1530 % can misbehave. That is the real cost of this default, and it is why the
1531 % escape exists.
1532 % * 'morewrites' still exists and now means "I insist": with it, a missing
1533 % package is an error rather than a warning.

```

```

1534 %
1535 % Under LuaTeX none of this applies and nothing is loaded.
1536 \newif\if@coppemorewrites \@coppemorewritestrue
1537 \newif\if@coppemwforced \@coppemwforcedfalse
1538 \DeclareOption{morewrites}{\@coppemorewritestrue\@coppemwforcedtrue}
1539 \DeclareOption{semmorewrites}{\@coppemorewritesfalse}
1540 \ProcessOptions\relax
1541 % morewrites has to come before anything else allocates, so it goes first
1542 % thing after the options are read.
1543 \if@coppemorewrites
1544   \ifdefined\directlua
1545     % Under LuaTeX there is nothing to solve: 128 streams already. The flag is
1546     % cleared too, so the \starttoc guard further down knows the ceiling is not
1547     % in play here either.
1548     \ClassInfo{coppe}{morewrites dispensado: o LuaTeX ja tem 128 fluxos}%
1549     \@coppemorewritesfalse
1550   \else
1551     \IfFileExists{morewrites.sty}
1552     {\RequirePackage{morewrites}}
1553     {\if@coppemwforced
1554       \ClassError{coppe}{Opcao 'morewrites' pedida, mas morewrites.sty nao
1555         esta instalado}{Instale o pacote morewrites, ou compile com lualatex,
1556         que nao precisa dele.}%
1557       \else
1558         \ClassWarningNoLine{coppe}{0 pacote morewrites nao esta instalado.\MessageBreak
1559           Sem ele o pdfTeX so aguenta 16 fluxos de escrita, e um trabalho\MessageBreak
1560           com todas as listas passa disso. Se a compilacao parar por falta\MessageBreak
1561           de fluxo, instale morewrites ou compile com lualatex}%
1562         \@coppemorewritesfalse
1563       \fi}%
1564   \fi
1565 \fi
1566 % \LoadClass and geometry ran before \ProcessOptions (see above), so the
1567 % two-sided layout can only be applied here. geometry reads left/right as
1568 % inner/outer once twoside is on, so the 3cm/2cm pair set above becomes the
1569 % mirrored gutter with no further arithmetic. \@twosidetrue is what book's own
1570 % twoside option sets, and is what \cleardoublepage and fancyhdr test.
1571 % In one-sided mode fancyhdr treats every page as odd, so the \fancyhead[R0,LE]
1572 % declarations elsewhere already put the folio in the top right corner of every
1573 % folha, as 2.7 requires.
1574 \if@coppetwoside
1575   \@twosidetrue
1576   \geometry{twoside}%
1577 \fi
1578 % Built-in language names (used by the loader and the dispatcher below).
1579 \def\coppe@@brazilianstr{brazilian}
1580 \def\coppe@@englishstr{english}
1581 % Load Babel with the foreign language first and the main language last
1582 % (Babel treats the LAST option in the list as the main language). With
1583 % pt-main this produces [english,brazilian]{babel} and with en-main
1584 % [brazilian,english] -- identical to the historical \if@english calls.
1585 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1586 \expandafter\RequirePackage\expandafter[\coppe@babelopts]{babel}
1587 % --- Multilingual string dispatcher -----

```

```

1588 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1589 % language <lang>. The three retrievers below look strings up either at the
1590 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1591 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1592 \providecommand{\copperdefstring}[3]{%
1593   \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1594 \providecommand{\coppestring}[1]{\@nameuse{coppe@str\languagename @#1}}
1595 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@coppe@mainlang @#1}}
1596 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@coppe@foreignlang @#1}}
1597 % Load an extra language pack at any point in the preamble (third-language
1598 % slot for citations or quotations beyond the main+foreign pair).
1599 \providecommand{\usecoppelanguage}[1]{%
1600   \def\coppe@@cur{#1}%
1601   \ifx\coppe@@cur\coppe@@brazilianstr\else
1602     \ifx\coppe@@cur\coppe@@englishstr\else
1603       \InputIfFileExists{coppe-lang-#1.def}{}
1604       {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1605     \fi\fi}
1606 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1607 % consulted by the cover/folha renderers once they are migrated in step 2.
1608 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1609 % still drive every visible site, so behavior for pt/en is unchanged.
1610 \providecommand{\titlein}[2]{%
1611   \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1612 % --- Built-in string tables: brazilian + english. The values mirror the
1613 % literals currently scattered through the class; step-1 only populates the
1614 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1615 \copperdefstring{brazilian}{appendix}{\Ap\~endice}
1616 \copperdefstring{english}{appendix}{Appendix}
1617 \copperdefstring{brazilian}{annex}{Anexo}
1618 \copperdefstring{english}{annex}{Annex}
1619 \copperdefstring{brazilian}{frame}{Quadro}
1620 \copperdefstring{english}{frame}{Frame}
1621 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1622 \copperdefstring{english}{listframe}{List of Frames}
1623 \copperdefstring{brazilian}{source}{Fonte}
1624 \copperdefstring{english}{source}{Source}
1625 \copperdefstring{brazilian}{program}{Programa}
1626 \copperdefstring{english}{program}{Program}
1627 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1628 \copperdefstring{english}{listprogram}{List of Programs}
1629 \copperdefstring{brazilian}{references}{Refer\~encias}
1630 \copperdefstring{english}{references}{References}
1631 \copperdefstring{brazilian}{in:}{em}
1632 \copperdefstring{english}{in:}{in}
1633 \copperdefstring{brazilian}{art}{Ilustra\c{c}\~ao}
1634 \copperdefstring{english}{art}{Art}
1635 \copperdefstring{brazilian}{platform}{Plataforma}
1636 \copperdefstring{english}{platform}{Platform}
1637 \copperdefstring{brazilian}{directedby}{Dire\c{c}\~ao}
1638 \copperdefstring{english}{directedby}{Directed by}
1639 \copperdefstring{brazilian}{producedby}{Produ\c{c}\~ao}
1640 \copperdefstring{english}{producedby}{Produced by}
1641 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}

```

```

1642 \copperdefstring{english} {listabbreviation}{List of Abbreviations and Acronyms}
1643 \copperdefstring{brazilian}{listsymbol} {Lista de S{\` i}mbolos}
1644 \copperdefstring{english} {listsymbol} {List of Symbols}
1645 \copperdefstring{brazilian}{glossary} {Gloss{\` a}rio}
1646 \copperdefstring{english} {glossary} {Glossary}
1647 \copperdefstring{brazilian}{advisor} {Orientador}
1648 \copperdefstring{english} {advisor} {Advisor}
1649 \copperdefstring{brazilian}{advisors} {Orientadores}
1650 \copperdefstring{english} {advisors} {Advisors}
1651 \copperdefstring{brazilian}{dept} {Programa}
1652 \copperdefstring{english} {dept} {Department}
1653 \copperdefstring{brazilian}{approvedmale} {Aprovado por}
1654 \copperdefstring{brazilian}{approvedfemale}{Aprovada por}
1655 \copperdefstring{brazilian}{approvedonmale} {Aprovado em}
1656 \copperdefstring{brazilian}{approvedonfemale}{Aprovada em}
1657 \copperdefstring{brazilian}{keywordsname} {Palavras-chave}
1658 \copperdefstring{brazilian}{volumename} {Volume}
1659 \copperdefstring{brazilian}{coadvisor} {Coorientador}
1660 \copperdefstring{brazilian}{coadvisors} {Coorientadores}
1661 \copperdefstring{brazilian}{ofname} {de}
1662 \copperdefstring{english} {approvedmale} {Approved by}
1663 \copperdefstring{english} {approvedfemale}{Approved by}
1664 \copperdefstring{english} {approvedonmale} {Approved on}
1665 \copperdefstring{english} {approvedonfemale}{Approved on}
1666 \copperdefstring{english} {keywordsname} {Keywords}
1667 \copperdefstring{english} {volumename} {Volume}
1668 \copperdefstring{english} {coadvisor} {Co-advisor}
1669 \copperdefstring{english} {coadvisors} {Co-advisors}
1670 \copperdefstring{english} {ofname} {of}
1671 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1672 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1673 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}
1674 \copperdefstring{brazilian}{cityname} {Rio de Janeiro}
1675 \copperdefstring{english} {cityname} {Rio de Janeiro}
1676 \copperdefstring{brazilian}{statename} {RJ}
1677 \copperdefstring{english} {statename} {RJ}
1678 \copperdefstring{brazilian}{countryname} {Brasil}
1679 \copperdefstring{english} {countryname} {Brasil}
1680 % Month names.
1681 \copperdefstring{brazilian}{monthname1} {Janeiro}
1682 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1683 \copperdefstring{brazilian}{monthname3} {Mar{\` c}o}
1684 \copperdefstring{brazilian}{monthname4} {Abril}
1685 \copperdefstring{brazilian}{monthname5} {Maio}
1686 \copperdefstring{brazilian}{monthname6} {Junho}
1687 \copperdefstring{brazilian}{monthname7} {Julho}
1688 \copperdefstring{brazilian}{monthname8} {Agosto}
1689 \copperdefstring{brazilian}{monthname9} {Setembro}
1690 \copperdefstring{brazilian}{monthname10}{Outubro}
1691 \copperdefstring{brazilian}{monthname11}{Novembro}
1692 \copperdefstring{brazilian}{monthname12}{Dezembro}
1693 \copperdefstring{english}{monthname1} {January}
1694 \copperdefstring{english}{monthname2} {February}
1695 \copperdefstring{english}{monthname3} {March}

```

```

1696 \copperdefstring{english}{monthname4} {April}
1697 \copperdefstring{english}{monthname5} {May}
1698 \copperdefstring{english}{monthname6} {June}
1699 \copperdefstring{english}{monthname7} {July}
1700 \copperdefstring{english}{monthname8} {August}
1701 \copperdefstring{english}{monthname9} {September}
1702 \copperdefstring{english}{monthname10}{October}
1703 \copperdefstring{english}{monthname11}{November}
1704 \copperdefstring{english}{monthname12}{December}
1705 % algorithm2e package option for this language (empty -> english default).
1706 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1707 \copperdefstring{english} {algorithm2eopt}{}
1708 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1709 \copperdefstring{brazilian}{babelname}{brazilian}
1710 \copperdefstring{english} {babelname}{english}
1711 % Auto-load language packs for any non-built-in language listed in babelopts.
1712 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1713 % (defined just above) and need no extra file; others (spanish/french/italian
1714 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1715 % must already be defined when these run, so the loader sits AFTER the tables.
1716 \@for\coppe@@lang:=\coppe@babelopts\do{%
1717   \edef\coppe@@cur{\coppe@@lang}%
1718   \ifx\coppe@@cur\coppe@@brazilianstr\else
1719   \ifx\coppe@@cur\coppe@@englishstr\else
1720     \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1721     {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1722     {\ClassError{coppe}
1723       {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1724       {Install it next to coppe.cls, or pick a built-in language.}}%
1725   \fi\fi}
1726 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1727 % Portuguese alias for short, inline quotations (proposta P3).
1728 \RequirePackage{csquotes}
1729 \newcommand{\citacao}[1]{\enquote{#1}}
1730 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
1731 % package gives the "Programa" listing (caption on top, ABNT R46) with
1732 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1733 % the language styles below are based on the user's listagens.tex. -----
1734 \RequirePackage{xcolor}
1735 \RequirePackage{listings}
1736 \RequirePackage{fancyvrb}
1737 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
1738 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
1739 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1740   literate=
1741   {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{a}}1 {\~}{\~{e}}1%
1742   {\~}{\~{e}}1 {\~}{\~{e}}1 {\~}{\~{i}}1 {\~}{\~{i}}1 {\~}{\~{o}}1%
1743   {\~}{\~{o}}1 {\~}{\~{o}}1 {\~}{\~{u}}1 {\~}{\~{u}}1 {\~}{\~{c}}1%
1744   {\~}{\~{c}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{A}}1 {\~}{\~{E}}1%
1745   {\~}{\~{E}}1 {\~}{\~{I}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{O}}1 {\~}{\~{O}}1%
1746   {\~}{\~{U}}1%
1747 }
1748 % Babel-aware listing names, set after listings' own babel setup.
1749 \AtBeginDocument{%

```



```

1750 \renewcommand\lstlistingname{\coppestring{program}}%
1751 \renewcommand\lstlistlistingname{\coppestring{listprogram}}%
1752 % \listofprogramas in the coppe list style (heading + ToC + running head).
1753 \newcommand\listofprogramas{%
1754   \chapter*{\lstlistlistingname}%
1755   \coppe@listtoc{\lstlistlistingname}%
1756   \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1757   \@starttoc{lol}}
1758 % Per-language styles + environments (python, java, xml, html, prolog).
1759 % The red hook that marks a wrapped line is built ONCE, into a box, and the
1760 % styles below insert the box. Writing the colour inline -- postbreak=\mbox{%
1761 % \textcolor{red}{\hookrightarrow}\space} -- worked for years and broke the
1762 % moment the pdfa option was switched on: pdfx puts xcolor into a converting
1763 % mode (\selectcolormodel plus \convertcolorsU/D), and a colour command run
1764 % from inside listings' line-breaking machinery then fails with "Missing
1765 % number, treated as zero" at the first wrapped line. Every listing in the
1766 % document dies, so the class's own example could not be compiled as PDF/A at
1767 % all. A box is immune: it is built here, in ordinary horizontal mode where
1768 % xcolor works normally, and listings only ever copies it.
1769 %
1770 % The \sbox waits for \begin{document} because xcolor's colours are not
1771 % defined while the class is still loading.
1772 \newsavebox\coppe@lstarrow
1773 \AtBeginDocument{\sbox\coppe@lstarrow{\textcolor{red}{\hookrightarrow}\space}}
1774 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1775   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1776   morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1777   emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1778   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1779   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1780   escapechar=',postbreak=\usebox\coppe@lstarrow}}
1781 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
1782   \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}%
1783 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!}%
1784 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}]}
1785 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1786   numberstyle=\small\ttfamily,numbersep=0.5em,%
1787   morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1788   keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1789   emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1790   emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1791   showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1792   postbreak=\usebox\coppe@lstarrow}}
1793 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{%
1794 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1795   numberstyle=\small\ttfamily,numbersep=0.5em,%
1796   keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1797   stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1798   extendedchars=true,inputencoding=utf8,breaklines=true,%
1799   postbreak=\usebox\coppe@lstarrow}}
1800 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{%
1801 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
1802   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1803   keywordstyle=\ttb\color{rgb}{0,0,0.5},emph={MyClass,__init__},%

```

```

1804 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
1805 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1806 breaklines=true,escapechar=',%
1807 postbreak=\usebox\coppe@lstarrow}}
1808 \lstnewenvironment{prolog}[1][\{\prologstyle\lstset{#1}%
1809 \csname\@lst @SetFirstNumber\endcsname\}{\csname\@lst @SaveFirstNumber\endcsname}
1810 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1811 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1812 morekeywords={self,as},keywordstyle=\ttb\color[rgb]{0,0,0.5},%
1813 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
1814 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1815 breaklines=true,escapechar=',%
1816 postbreak=\usebox\coppe@lstarrow}}
1817 \lstnewenvironment{java}[1][\{\javastyle\lstset{#1}%
1818 \csname\@lst @SetFirstNumber\endcsname\}{\csname\@lst @SaveFirstNumber\endcsname}
1819 \newcommand\javainline[1][\{\javastyle\lstinline!#1!}%
1820 \newcommand\javaexternal[2][\{\javastyle\lstinputlisting[1]{#2}}%
1821 % fancyvrb's Verbatim environment is available for program output; the
1822 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1823 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1824 % example,
1825 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1826 % ...
1827 % \end{Verbatim}
1828 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1829 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
1830 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1831 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1832 % language (English) -- needed because algorithm2e errors on an empty option.
1833 \edef\coppe@@algotlang{\coppemainstring{algorithm2eopt}}
1834 \ifx\coppe@@algotlang\@empty
1835 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1836 \else
1837 \expandafter\RequirePackage\expandafter[\coppe@@algotlang,linesnumbered,lined,algochapter,r
1838 \fi
1839 % algorithm2e composes its own caption and therefore ignores the class-wide
1840 % \captionsetup{font=footnotesize,labelsep=endash}. Left alone it printed
1841 % "Algoritmo 6.1:Titulo" in bold at body size, while every other illustration
1842 % printed "Figura 6.1 -- Titulo" in footnotesize -- 2.10 asks for the
1843 % designating word, the number, a TRAVESSAO and the title, uniformly.
1844 %
1845 % The separator and the caption fonts are therefore set to match. The 'ruled'
1846 % layout (rules above and below) is kept: 2.10 prescribes where the caption and
1847 % the source go, not whether the float is delimited, and the rules read here
1848 % like the top-and-bottom delimitation a table carries.
1849 \SetAlgoCaptionSeparator{\textendash}
1850 \SetAlCapFnt{\footnotesize}
1851 \SetAlCapNameFnt{\footnotesize}
1852 \SetAlCapSty{textnormal}
1853 \SetAlCapNameSty{textnormal}
1854 \SetKwFor{Para}{para}{faça}{fim do para}
1855 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1856 % setspace + hyperref + biblatex are loaded at the very end of the class
1857 % (see the block just before %</class>), so their hooks see every macro the

```

```

1858 % class defines, and the load order setspace -> hyperref -> biblatex is
1859 % respected.

\department This macro is used to set the author's affiliation. There are twelve options which
              correspond to all academic units at COPPE/UFRJ. It defines the current and the
              foreign names of these units.

1860 \newcommand\coppe@deptcode{}
1861 \newcommand\meta@deptname{}
1862 \newcommand\department[1]{%
1863 % The programme CODE is kept alongside the resolved name, and each name is
1864 % declared TWICE. \local@deptname keeps the TeX accent escapes with braces
1865 % (Computa{\c c}{\~ a}o) that the cover and the folha de rosto typeset;
1866 % \meta@deptname is the same name in literal UTF-8, and is the one that goes
1867 % into the XMP. pdfx copies whatever it is given straight through, so feeding it
1868 % the escaped form produced "Computa{c}o" in dc:description -- which is why the
1869 % subject line used to fall back to the bare programme code. Both forms are
1870 % written out here rather than derived from one another because there is no
1871 % reliable way to strip TeX accent syntax at the point where the .xmpdata is
1872 % written, and a table of thirteen entries is cheaper than that machinery.
1873 \gdef\coppe@deptcode{#1}%
1874 \ifthenelse{\equal{#1}{PEB}}{
1875   {\global\def\local@deptname{Engenharia Biom{\` e}dica}
1876   \global\def\meta@deptname{Engenharia Biomédica}
1877   \global\def\foreign@deptname{Biomedical Engineering}}{}
1878 \ifthenelse{\equal{#1}{PEC}}{
1879   {\global\def\local@deptname{Engenharia Civil}
1880   \global\def\meta@deptname{Engenharia Civil}
1881   \global\def\foreign@deptname{Civil Engineering}}{}
1882 \ifthenelse{\equal{#1}{PEE}}{
1883   {\global\def\local@deptname{Engenharia El{\` e}trica}
1884   \global\def\meta@deptname{Engenharia Elétrica}
1885   \global\def\foreign@deptname{Electrical Engineering}}{}
1886 \ifthenelse{\equal{#1}{PEM}}{
1887   {\global\def\local@deptname{Engenharia Mec{\` a}nica}
1888   \global\def\meta@deptname{Engenharia Mecânica}
1889   \global\def\foreign@deptname{Mechanical Engineering}}{}
1890 \ifthenelse{\equal{#1}{PEMM}}{
1891   {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
1892   \global\def\meta@deptname{Engenharia Metalúrgica e de Materiais}
1893 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
1894 \ifthenelse{\equal{#1}{PEN}}{
1895   {\global\def\local@deptname{Engenharia Nuclear}
1896   \global\def\meta@deptname{Engenharia Nuclear}
1897   \global\def\foreign@deptname{Nuclear Engineering}}{}
1898 \ifthenelse{\equal{#1}{PEN0}}{
1899   {\global\def\local@deptname{Engenharia Oce{\` a}nica}
1900   \global\def\meta@deptname{Engenharia Oceânica}
1901   \global\def\foreign@deptname{Ocean Engineering}}{}
1902 \ifthenelse{\equal{#1}{PPE}}{
1903   {\global\def\local@deptname{Planejamento Energ{\` e}tico}
1904   \global\def\meta@deptname{Planejamento Energético}
1905   \global\def\foreign@deptname{Energy Planning}}{}
1906 \ifthenelse{\equal{#1}{PEP}}{
1907   {\global\def\local@deptname{Engenharia de Produ{\c c}{\~ a}o}

```

```

1908 \global\def\meta@deptname{Engenharia de Produção}
1909 \global\def\foreign@deptname{Production Engineering}}{}
1910 \ifthenelse{\equal{#1}{PEQ}}
1911 {\global\def\local@deptname{Engenharia Qu\` i mica}
1912 \global\def\meta@deptname{Engenharia Química}
1913 \global\def\foreign@deptname{Chemical Engineering}}{}
1914 \ifthenelse{\equal{#1}{PESC}}
1915 {\global\def\local@deptname{Engenharia de Sistemas e Computa\c c}{\~ a o}
1916 \global\def\meta@deptname{Engenharia de Sistemas e Computação}
1917 \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1918 \ifthenelse{\equal{#1}{PET}}
1919 {\global\def\local@deptname{Engenharia de Transportes}
1920 \global\def\meta@deptname{Engenharia de Transportes}
1921 \global\def\foreign@deptname{Transportation Engineering}}{}
1922 \ifthenelse{\equal{#1}{PENT}}
1923 {\global\def\local@deptname{Engenharia de Nanotecnologia}
1924 \global\def\meta@deptname{Engenharia de Nanotecnologia}
1925 \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1926 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

1927 \renewcommand\title[1]{%
1928 \global\def\local@title{#1}%
1929 \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
1930 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

1931 \newcommand\foreigntitle[1]{%
1932 \global\def\foreign@title{#1}%
1933 \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
1934 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```

1935 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
1936 % Subtitle, mirroring the title machinery slot for slot: \subtitle feeds the
1937 % brazilian slot, \foreignsubtitle the english one, \subtitlein any other.
1938 % 3.1.1 and 3.1.2.1.1(c) require the subtitle on the capa and on the folha de
1939 % rosto, subordinated to the title by a colon.
1940 \newcommand\subtitle[1]{%
1941 \expandafter\gdef\csname coppe@subtitle@brazilian\endcsname{#1}}
1942 \newcommand\foreignsubtitle[1]{%
1943 \expandafter\gdef\csname coppe@subtitle@english\endcsname{#1}}
1944 \newcommand\subtitlein[2]{%
1945 \expandafter\gdef\csname coppe@subtitle@#1\endcsname{#2}}
1946 \newcommand\coppe@selectsubtitle{\@nameuse{coppe@subtitle@\coppe@mainlang}}
1947 % Title as the capa and the folha de rosto print it. Annexes A and B show the
1948 % title in caps and the subtitle in ordinary case after a colon, so the case
1949 % change is applied to the title alone.

```

```

1950 \newcommand\coppe@titleblock{%
1951   \MakeUppercase{\coppe@selecttitle}%
1952   \@ifundefined{coppe@subtitle@\coppe@mainlang}{\coppe@selectsubtitle}}
1953 % Number of volumes (3.1.1 and 3.1.2.1.1(d)): printed only when there is more
1954 % than one, since a single-volume work says nothing about volumes.
1955 % Coorientador (3.1.2.1.1(f)). Same shape as \advisor, including the optional
1956 % institution; the label switches to the plural on the second call, as the
1957 % advisor label already did.
1958 \newcount\@coadvisor\@coadvisor0
1959 \newcommand\coadvisor[5][\coppe@titleblock]{%
1960   \global\@namedef{CoppeCoadvisorInst:\expandafter\the\@coadvisor}{#1}
1961   \global\@namedef{CoppeCoadvisorTitle:\expandafter\the\@coadvisor}{#2}
1962   \global\@namedef{CoppeCoadvisorName:\expandafter\the\@coadvisor}{#3}
1963   \global\@namedef{CoppeCoadvisorSurname:\expandafter\the\@coadvisor}{#4}
1964   \global\@namedef{CoppeCoadvisorDegree:\expandafter\the\@coadvisor}{#5}
1965   \global\advance\@coadvisor by 1
1966 }
1967 \newcommand\local@coadvisorstring{%
1968   \ifnum\@coadvisor>1
1969     \coppemainstring{coadvisors}%
1970   \else
1971     \coppemainstring{coadvisor}%
1972   \fi}
1973 % Mesma coisa no idioma estrangeiro, para a pagina de abstract.
1974 \newcommand\foreign@coadvisorstring{%
1975   \ifnum\@coadvisor>1
1976     \coppeforeignstring{coadvisors}%
1977   \else
1978     \coppeforeignstring{coadvisor}%
1979   \fi}
1980 \newcommand\coppe@nvolumes{1}
1981 \newcommand\coppe@thisvolume{1}
1982 \newcommand\volumes[1]{\gdef\coppe@nvolumes{#1}}
1983 \newcommand\volume[1]{\gdef\coppe@thisvolume{#1}}
1984 \newcommand\coppe@volumeline{%
1985   \ifnum\coppe@nvolumes>1
1986     \par\vspace*{4mm}%
1987     \coppemainstring{volumename}\space\coppe@thisvolume\space
1988     \coppemainstring{ofname}\space\coppe@nvolumes
1989 % The closing \par matters: on the capa the next thing is \vspace{\stretch{3}},
1990 % which does not end a paragraph, so without it the city ran on into the volume
1991 % line ("Volume 2 de 3 RIO DE JANEIRO").
1992   \par
1993   \fi}

```

\advisor Defines globally the title, name and academic degree of the advisors.

```

1994 \newcount\@advisor\@advisor0
1995 % The optional first argument is the member's institution, required by
1996 % 3.1.2.1.3(e). Omitting it keeps every pre-4.1 call working unchanged.
1997 \newcommand\advisor[5][\coppe@titleblock]{%
1998   \global\@namedef{CoppeAdvisorInst:\expandafter\the\@advisor}{#1}
1999   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#2}
2000   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#3}
2001   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#4}

```

```

2002 \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#5}
2003 \global\advance\@advisor by 1
2004 \ifnum\@advisor>1
2005 \renewcommand\local@advisorstring{Orientadores}
2006 \renewcommand\foreign@advisorstring{Advisors}
2007 \fi
2008 }

```

`\examiner`

```

2009 \newcount\@examiner\@examiner0
2010 % Until v4.1 this stored only "#1 #2" and DISCARDED the third argument: the
2011 % titulacao every document passed was silently thrown away and never printed,
2012 % though 3.1.2.1.3(e) requires it. Each part is now kept separately, and the
2013 % optional first argument carries the institution.
2014 % CoppeExaminer: is retained with its old meaning for anything reading it.
2015 \newcommand\examiner[4][\%
2016 \global\@namedef{CoppeExaminerInst:\expandafter\the\@examiner}{#1}
2017 \global\@namedef{CoppeExaminerTitle:\expandafter\the\@examiner}{#2}
2018 \global\@namedef{CoppeExaminerName:\expandafter\the\@examiner}{#3}
2019 \global\@namedef{CoppeExaminerDegree:\expandafter\the\@examiner}{#4}
2020 \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#2\ #3}
2021 \global\advance\@examiner by 1
2022 }

```

`\author` It was redefined to allow the identification of the author's first names and surname.

```

2023 \renewcommand\author[2]{%
2024 \global\def\@authname{#1}
2025 \global\def\@authsurn{#2}
2026 }

```

`\date` Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (`\coppe@docmonth` / `\coppe@docyear`) rather than mutating TeX's `\month` / `\year` primitives, which would clobber `\today`.

```

2027 \renewcommand\date[2]{%
2028 \def\coppe@docmonth{#1}%
2029 \def\coppe@docyear{#2}}

```

`\local@monthname`

```

2030 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
2031 Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
2032 Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}

```

`\foreign@monthname`

```

2033 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
2034 January\or February\or March\or April\or May\or June\or
2035 July\or August\or September\or October\or November\or December\fi}

```

`\keyword`

```

2036 \newcounter{keywords}
2037 \newcommand\keyword[1]{%
2038 \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}

```

```

2039 \global\addtocounter{keywords}{1}
2040 }
2041 % Keywords for the foreign abstract. Annex F of the manual shows the English
2042 % abstract closing with English keywords, and \keyword alone cannot serve both
2043 % pages. When no \foreignkeyword is given the foreign abstract falls back to
2044 % the \keyword list, which is what a document written before v4.1 expects.
2045 \newcounter{foreignkeywords}
2046 \newcommand\foreignkeyword[1]{%
2047   \global\@namedef{CoppeForeignKeyword:\expandafter\the\c@foreignkeywords}{#1}
2048   \global\addtocounter{foreignkeywords}{1}
2049 }
2050 % Keywords for the optional third abstract (Portuguese, for the banca). Same
2051 % shape and same fallback as \foreignkeyword: without it the Portuguese page
2052 % would close with the main-language keywords under a Portuguese label, which
2053 % is what a Spanish-, French- or Italian-main work would produce.
2054 \newcounter{braziliankeywords}
2055 \newcommand\braziliankeyword[1]{%
2056   \global\@namedef{CoppeBrazilianKeyword:\expandafter\the\c@braziliankeywords}{#1}
2057   \global\addtocounter{braziliankeywords}{1}
2058 }
2059 % 3.1.2.1.4: the keywords close the resumo, preceded by the term, a colon, and
2060 % separated from one another by semicolons.
2061 % #1 label, #2 csname prefix, #3 how many.
2062 \newcommand\coppe@printkeywords[3]{%
2063   \ifnum#3>0
2064     \par\vspace*{7mm}%
2065     \noindent{\singlespacing\textbf{#1:}\enspace
2066       \count1=0
2067       \@whilenum\count1<#3\do{%
2068         \ifnum\count1>0 ;\space\fi
2069         \csname #2\the\count1\endcsname
2070         \advance\count1 by 1}%
2071       .\par}%
2072   \fi}

\areaconcentracao Data for the folha adicional (Coleta CAPES) and for the natureza block of the
\linhapesquisa folha de rosto and folha de aprovacao. Every one of them defaults to empty;
\tipoproducao \makefolhaadicional prints an empty rule where a value is missing, so the pro-
\projetovinculado gramme can complete the sheet by hand if need be.
\nomeprojeto 2073 \newcommand\coppe@areaconc{}
\agenciafomento 2074 \newcommand\areaconcentracao[1]{\gdef\coppe@areaconc{#1}}
2075 \newcommand\coppe@dataaprov{}
2076 \newcommand\dataaprovacao[1]{\gdef\coppe@dataaprov{#1}}
2077 \newcommand\coppe@linhapesq{}
2078 \newcommand\linhapesquisa[1]{\gdef\coppe@linhapesq{#1}}
2079 \newcommand\coppe@tipoproducao{}
2080 \newcommand\tipoproducao[1]{\gdef\coppe@tipoproducao{#1}}
2081 \newcommand\coppe@projetovinc{}
2082 \newcommand\projetovinculado[1]{\gdef\coppe@projetovinc{#1}}
2083 \newcommand\coppe@nomeprojeto{}
2084 \newcommand\nomeprojeto[1]{\gdef\coppe@nomeprojeto{#1}}
2085 \newcounter{coppeagencias}
2086 \newcommand\agenciafomento[2]{%
2087   \global\@namedef{CoppeAgencia:\expandafter\the\c@coppeagencias}{#1}%

```

```

2088 \global\@namedef{CoppeAgenciaSigla:\expandafter\the\c@coppeagencias}{#2}%
2089 \global\addtocounter{coppeagencias}{1}}

```

`\freeconfig` This command allows easy changing of core class parameters.

```

2090 \newcommand\freeconfig[6]{%
2091 \providecommand\@degree{}
2092 \renewcommand\@degree{#1}
2093 \providecommand\@degreename{}
2094 \renewcommand\@degreename{#2}
2095 \providecommand\local@degname{}
2096 \renewcommand\local@degname{#3}
2097 \providecommand\foreign@degname{}
2098 \renewcommand\foreign@degname{#4}
2099 \providecommand\local@doctype{}
2100 \renewcommand\local@doctype{#5}
2101 \providecommand\foreign@doctype{}
2102 \renewcommand\foreign@doctype{#6}%
2103 }
2104 % \end{macrocode}
2105 % \end{macro}
2106 %
2107 % \begin{macro}{\frontmatter}
2108 % The number of pages for both frontmatter and mainmatter printed
2109 % in the cataloging details page is computed by means of simple
2110 % \LaTeX\ labels.
2111 % \begin{macrocode}
2112 \renewcommand\frontmatter{%
2113 \cleardoublepage
2114 \@mainmatterfalse
2115 % The count already started at the folha de rosto (see \maketitle) and simply
2116 % continues here: 2.7 requires every folha from the folha de rosto onwards to
2117 % be counted sequentially, printing arabic numerals only from the first textual
2118 % folha. Resetting the counter here -- as this class did until v4.1 -- made the
2119 % Introducao come out as page 1, which the norm does not allow.
2120 %
2121 % The numeraisromanos option keeps the legacy behaviour (lowercase roman in the
2122 % pre-textual part, arabic restarting at 1 in the textual part). That departs
2123 % from 2.7, which has the pre-textual folhas counted but NOT numbered, so the
2124 % option warns.
2125 \if@copperoman
2126 \ClassWarningNoLine{coppe}{Option ‘numeraisromanos’ prints roman numerals
2127 on the pre-textual folhas and restarts the arabic count at 1 in the
2128 textual part. Section 2.7 of the SiBI manual (9th rev. ed., 2026) has
2129 those folhas counted but not numbered, and the arabic numbering
2130 continuing the count. Drop the option to comply}%
2131 \pagenumbering{roman}%
2132 \fi
2133 \thispagestyle{empty}
2134 \makefrontpage
2135 \clearpage
2136 \if@copperoman
2137 \fancypagestyle{plain}{\fancyhf{} \fancyhead[R0,LE]{\coppe@folio}%
2138 \renewcommand{\headrulewidth}{0pt}}%
2139 \pagestyle{coppe}%

```



```

2140 \else
2141   \fancypagestyle{plain}{\fancyhf{}}\renewcommand{\headrulewidth}{0pt}}%
2142   \pagestyle{empty}%
2143 \fi
2144 }

\mainmatter
2145 \renewcommand\mainmatter{%
2146   \coppe@mainBegin
2147   \cleardoublepage
2148   \@mainmattertrue
2149   \fancypagestyle{plain}{\fancyhf{}}\fancyhead[R0,LE]{\coppe@folio}%
2150   \renewcommand{\headrulewidth}{0pt}}%
2151   \pagestyle{coppe}%
2152 % No \pagenumbering{arabic} in the default path: it would reset the counter to
2153 % 1 and make the Introducao folha 1, contrary to 2.7. The counter has been
2154 % arabic and running since the folha de rosto; all that changes here is that
2155 % the folio starts being PRINTED. Only the legacy numeraisromanos path, which
2156 % switched to roman in the front matter, has to switch back -- and there the
2157 % restart at 1 is intrinsic to that (non-conforming) scheme.
2158 \if@copperoman\pagenumbering{arabic}\fi}

```

\backmatter

```

2159 \renewcommand\backmatter{%
2160   \if@openright
2161     \cleardoublepage
2162   \else
2163     \clearpage
2164   \fi}
2165 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```

2166 \providecommand{\coppefinalengine}{%
2167   \ifPDFTeX\hologo{pdfLaTeX}\fi
2168   \ifLuaTeX\hologo{LuaLaTeX}\fi
2169   \ifXeTeX\hologo{XeLaTeX}\fi}
2170 \providecommand{\coppefinalfont}{Computer Modern (padr\~ao \LaTeX{})}
2171 \providecommand{\coppefinalmanual}{%
2172   \emph{Norma para a Elaborac\~ao Gr\~afica de Teses e Disserta\~oes
2173     da COPPE/UFRJ}}
2174 \newcommand{\coppetexfinalpage}{%
2175   \clearpage

```

```

2176 % Force the colophon onto a verso (even) page so it faces the rear cover.
2177 % If the current page is odd (recto), fill it with an empty verso first.
2178 \if@twoside
2179     \ifodd\c@page
2180         \hbox{}\thispagestyle{empty}\newpage
2181     \fi
2182 \fi
2183 \thispagestyle{empty}%
2184 \null\vfill
2185 \begin{otherlanguage}{brazilian}%
2186     \begin{center}
2187         {\footnotesize\setstretch{1.0}}%
2188         \textsc{Colof\~ao}\par
2189         \vspace{0.7em}
2190         \begin{minipage}{0.75\textwidth}\centering
2191             Este documento foi composto em \LaTeX{} com \coppefinalengine{}
2192             em \today, na fonte \coppefinalfont. A bibliografia foi processada
2193             por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
2194             vers\~ao\coppe@version, dispon\~i vel no reposit\~orio GitHub
2195             \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
2196             conformidade com a \coppefinalmanual.
2197         \end{minipage}\par
2198     \end{center}
2199 \end{otherlanguage}%
2200 \vspace*{2cm}%
2201 \clearpage

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

2202 \newcommand\makecover{%
2203     \begin{titlepage}%
2204     \centering
2205     \makebox[\linewidth]{%
2206         \includegraphics[height=2cm]{ufrj-logo}\hfill
2207         \includegraphics[height=2cm]{coppe-logo}}\par
2208     \vspace{1.5cm}
2209     {\singlespacing
2210 \MakeUppercase{\local@universityname}\par
2211     \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \
2212         P\~o s-Gradua\~c o e Pesquisa de Engenharia}}\par
2213     \MakeUppercase{Programa de \local@deptname}\par}
2214     \vspace{\stretch{2}}
2215     \textbf{\MakeUppercase{\textbf{\@authname\ \@authsurn}}}\par
2216     \vspace{\stretch{2}}
2217     \nohyphens{\coppe@titleblock}\par
2218     \coppe@volumeline
2219     \vspace{\stretch{3}}
2220     \MakeUppercase{\local@cityname}\par
2221     \coppe@docyear

```

```

2222 \end{titlepage}%
2223 }

\maketitle

2224 \renewcommand\maketitle{%
2225 % The capa is neither counted nor numbered (2.7), so it lives in its own
2226 % throwaway numbering scheme; \pagenumbering{arabic} right after it restarts
2227 % the counter, making the FOLHA DE ROSTO page 1 -- which is where 2.7 says the
2228 % count begins. Nothing is printed here: \frontmatter keeps \pagestyle{empty}
2229 % until the textual part starts.
2230 \pagenumbering{alph}
2231 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
2232 \makecover
2233 \pagenumbering{arabic}
2234 \begin{titlepage}
2235 \begin{flushleft}
2236 \vspace*{1.5mm}
2237 \setlength\baselineskip{0pt}
2238 \setlength\parskip{1mm}
2239 \makebox[\linewidth]{%
2240 \includegraphics[height=2cm]{ufrj-logo}\hfill
2241 \includegraphics[height=2cm]{coppe-logo}}
2242 \end{flushleft}
2243 \vspace{1.05cm}
2244 \begin{center}
2245 \nohyphens{\@authname\ \@authsurn}\par
2246 \vspace*{3cm}
2247 \nohyphens{\coppe@titleblock}\par
2248 \coppe@volumeline
2249 \end{center}
2250 \vspace*{2.1cm}
2251 \begin{flushright}
2252 % 2.4: on the folha de rosto the natureza/objetivo/instituicao block is set
2253 % SINGLE spaced, and aligned from the middle of the mancha grafica to the right
2254 % margin. 0.5\textwidth tracks the mancha, so the block follows automatically if
2255 % the margins ever change -- the old fixed 8.45cm did not.
2256 \begin{minipage}{0.5\textwidth}
2257 \singlespacing\frontcover@maintext
2258 \end{minipage}\par
2259 \end{flushright}
2260 % Only the natureza block is indented to the right (2.4). The linha de pesquisa
2261 % and the orientadores return to the LEFT MARGIN, as the CAPES model of the
2262 % folha de rosto and Annex B of the manual show. Until v4.1 they sat inside the
2263 % right-hand block, which is what a recent review flagged.
2264 \vspace*{12mm}
2265 {\singlespacing
2266 \ifthenelse{\equal{\coppe@linhapesq}{}}{\}%
2267 \noindent Linha de pesquisa: \coppe@linhapesq\par
2268 \vspace*{5mm}}%
2269 \nohyphens{%
2270 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }}>\raggedright\arraybackslashX@{}}
2271 \local@advisorstring: &
2272 \count1=0
2273 \toks@={}

```

```

2274 \@whilenum \count1<\@advisor \do{%
2275 \ifcase\count1 % same as \ifnum0=\count1
2276 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2277 \expandafter\endcsname\expandafter\space%
2278 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2279 \else
2280 \toks@=\expandafter\expandafter\expandafter{%
2281 \expandafter\the\expandafter\toks@%
2282 \expandafter&\expandafter\space%
2283 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2284 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2285 }%
2286 \fi
2287 \advance\count1 by 1}
2288 \the\toks@
2289 \end{tabularx}}%
2290 \ifnum\@coadvisor>0
2291 \par\vspace*{2mm}%
2292 \nohyphens{%
2293 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2294 \local@coadvisorstring: &
2295 \count1=0
2296 \toks@={}
2297 \@whilenum \count1<\@coadvisor \do{%
2298 \ifcase\count1
2299 \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2300 \expandafter\endcsname\expandafter\space%
2301 \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2302 \else
2303 \toks@=\expandafter\expandafter\expandafter{%
2304 \expandafter\the\expandafter\toks@%
2305 \expandafter&\expandafter\space%
2306 \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2307 \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\
2308 }%
2309 \fi
2310 \advance\count1 by 1}
2311 \the\toks@
2312 \end{tabularx}}%
2313 \fi
2314 \par}
2315 \vspace*{\fill}
2316 % 3.1.2.1.1(g): the folha de rosto carries the ANO DE DEPOSITO alone -- the year
2317 % the revised final version is handed to the Programa. The month was never asked
2318 % for here (the capa, Annex A, also shows only the year), and printing it
2319 % invited confusion with the defence date. \date still takes month and year:
2320 % the month is what the abstract pages and the colophon use.
2321 \begin{center}
2322 \local@cityname\par
2323 \coppe@docyear
2324 \end{center}
2325 \end{titlepage}
2326 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
2327 % the title page is a recto, so this lands on the immediately following verso,

```

```

2328 % ahead of the approval sheet produced by \frontmatter.
2329 % LaTeX's titlepage environment ends with \setcounter{page}{1} whenever the
2330 % document is one-sided (see \endtitlepage in classes.dtx). That silently ate
2331 % the folha de rosto from the count, putting the folha de aprovacao at 1 instead
2332 % of 2. The count is therefore restated here: the folha de rosto WAS folha 1, so
2333 % whatever follows it is folha 2.
2334 \setcounter{page}{2}%
2335 % The page carrying the ficha catalografica is explicitly excluded from the
2336 % count -- 2.7 of the 2026 manual ("a folha adicional que contem a ficha
2337 % catalografica nao pode ser contada ou numerada"), and the 2025 edition said
2338 % the same of the verso of the folha de rosto. Hence the counter is stepped
2339 % back after it.
2340 \ifthenelse{\boolean{maledoc}}{\}%
2341 \makefolhaadicional\clearpage\addtocounter{page}{-1}%
2342 \global\let\maketitle\relax%
2343 \global\let\and\relax}

2344 % 3.1.2.1.3 lists the elements of the folha de aprovacao in order: (a) the
2345 % AUTHOR, (b) the title in full, (c) the natureza/objetivo/instituicao/area de
2346 % concentracao, (d) the date of approval, (e) the examining board. Until v4.1
2347 % this page opened with the title in caps and the author under it, and set the
2348 % natureza as a wall of uppercase (\frontpage@maintext). Annex D shows neither.
2349 % One board member. \coppe@membroid is the identification line required by
2350 % 3.1.2.1.3(e) -- name, titulacao and institution -- with the last two omitted
2351 % when not given, so a document that never supplied them looks as it did before.
2352 \newcommand\coppe@membroid[4]{%
2353   #1\ #2%
2354   \ifthenelse{\equal{#3}{}}{\},{ #3}%
2355   \ifthenelse{\equal{#4}{}}{\},{ #4}%
2356 % The gap above each signature rule is \coppe@membrogap, set from the size of
2357 % the board just before the list is assembled (see \makefrontpage). A fixed 9mm
2358 % fitted five members and overflowed at seven -- the sheet silently continued
2359 % onto a second page, which also printed a folio in the pre-textual part, where
2360 % 2.7 forbids one. The shrink lives in the length itself -- \newlength makes a
2361 % skip register, so "9mm minus 2mm" survives \setlength and reaches \vspace as
2362 % real glue, and a title long enough to wrap an extra line is absorbed instead
2363 % of costing a page. It has to go through \vspace and not a bare \vskip: the
2364 % board list is emitted inside \nohyphens, i.e. in horizontal mode, where
2365 % \vskip is an error and TeX then typesets the "plus ... minus ..." after it as
2366 % ordinary text -- which is exactly what landed on the page the first time.
2367 \newlength\coppe@membrogap
2368 \setlength\coppe@membrogap{9mm minus 2mm}
2369 \newcount\coppe@bancasize
2370 \newcommand\coppe@membro[4]{%
2371   \if@coppeassin
2372     \vspace{\coppe@membrogap}%
2373     \begin{center}
2374       \rule{95mm}{0.4pt}\par
2375       \nopagebreak{\footnotesize\coppe@membroid{#1}{#2}{#3}{#4}}\par
2376     \end{center}%
2377   \else
2378     \noindent\hspace*{8mm}%
2379     \begin{minipage}{\dimexpr\textwidth-8mm\relax}
2380       \coppe@membroid{#1}{#2}{#3}{#4}

```

```

2381 \end{minipage}\par\vspace{2mm}%
2382 \fi}
2383 \newcommand\makefrontpage{%
2384 \begin{center}
2385 \nohyphens{\@authname\ \@authsur}\par
2386 \vspace*{7mm}
2387 \sloppy\nohyphens{\coppe@selecttitle
2388 \@ifundefined{coppe@subtitle@\coppe@mainlang}{\{:
2389 \coppe@selectsubtitle}}\par
2390 \end{center}\par
2391 \vspace*{8mm}
2392 % Same block, same indentation and same single spacing as the folha de rosto:
2393 % 2.4 names the two pages together.
2394 \begin{flushright}
2395 \begin{minipage}{0.5\textwidth}
2396 \singlespacing\frontcover@maintext
2397 \end{minipage}
2398 \end{flushright}
2399 \vspace*{12mm}
2400 % (d) date of approval. Annex D prints "Aprovada em:" followed by a blank to be
2401 % filled in; \dataaprovacao supplies it when the date is already known.
2402 \noindent\local@approvedonname:\enspace
2403 \ifthenelse{\equal{\coppe@dataaprov}{}}{\hrulefill}{\coppe@dataaprov}\par
2404 \vspace*{12mm}
2405 % (e) the examining board, with the ORIENTADOR FIRST, as president of the board
2406 % (3.1.2.1.3(e)). Until v4.1 the advisors were printed in a block of their own
2407 % above a second list headed "Aprovada por:", which split the board in two and
2408 % did not mark the advisor as its president. They are now one list.
2409 %
2410 % The list is assembled into \coppe@bancalist with \g@addto@macro instead of
2411 % being built inside a tabularx: each entry now carries four fields (treatment,
2412 % name, titulacao, institution) and the old \toks@/\expandafter accumulation
2413 % could not carry them without becoming unreadable. \protected@edef keeps
2414 % accents and other protected commands in the names intact.
2415 % Size of the board decides the spacing above. The thresholds are measured, not
2416 % guessed: with the 9mm gap a seven-member board pushed three members onto a
2417 % second page, 7mm brought six back onto one sheet and 5mm did the same for
2418 % seven and eight. Boards of five or fewer keep the original 9mm, so every
2419 % document written before this change renders exactly as it did.
2420 \coppe@bancasize=\@advisor
2421 \advance\coppe@bancasize by \@coadvisor
2422 \advance\coppe@bancasize by \@examiner
2423 \ifnum\coppe@bancasize>6
2424 \setlength\coppe@membrogap{5mm minus 2mm}%
2425 \else\ifnum\coppe@bancasize>5
2426 \setlength\coppe@membrogap{7mm minus 2mm}%
2427 \fi\fi
2428 \gdef\coppe@bancalist{}%
2429 \count1=0
2430 \@whilenum\count1<\@advisor\do{%
2431 \protected@edef\coppe@tmp{\noexpand\coppe@membro
2432 {\csname CoppeAdvisorTitle:\the\count1\endcsname}}%
2433 {\csname CoppeAdvisorName:\the\count1\endcsname\space
2434 \csname CoppeAdvisorSurname:\the\count1\endcsname}}%

```

```

2435     {\csname CoppeAdvisorDegree:\the\count1\endcsname}%
2436     {\csname CoppeAdvisorInst:\the\count1\endcsname}}%
2437     \expandafter\g@addto@macro\expandafter\coppe@bancalist
2438     \expandafter{\coppe@tmp}%
2439     \advance\count1 by 1}%
2440 % Coorientadores come right after the orientadores and before the rest of the
2441 % board, keeping the order 3.1.2.1.3(e) prescribes.
2442 \count1=0
2443 \@whilenum\count1<\@coadvisor\do{%
2444     \protected@edef\coppe@tmp{\noexpand\coppe@membro
2445         {\csname CoppeCoadvisorTitle:\the\count1\endcsname}%
2446         {\csname CoppeCoadvisorName:\the\count1\endcsname}\space
2447         \csname CoppeCoadvisorSurname:\the\count1\endcsname}%
2448         {\csname CoppeCoadvisorDegree:\the\count1\endcsname}%
2449         {\csname CoppeCoadvisorInst:\the\count1\endcsname}}%
2450     \expandafter\g@addto@macro\expandafter\coppe@bancalist
2451     \expandafter{\coppe@tmp}%
2452     \advance\count1 by 1}%
2453 \count1=0
2454 \@whilenum\count1<\@examiner\do{%
2455     \protected@edef\coppe@tmp{\noexpand\coppe@membro
2456         {\csname CoppeExaminerTitle:\the\count1\endcsname}%
2457         {\csname CoppeExaminerName:\the\count1\endcsname}%
2458         {\csname CoppeExaminerDegree:\the\count1\endcsname}%
2459         {\csname CoppeExaminerInst:\the\count1\endcsname}}%
2460     \expandafter\g@addto@macro\expandafter\coppe@bancalist
2461     \expandafter{\coppe@tmp}%
2462     \advance\count1 by 1}%
2463 \noindent\local@approvedname:\par
2464 \nohyphens{\singlespacing\coppe@bancalist}\par
2465 % No city/date block at the foot: 3.1.2.1.3 ends the folha de aprovacao at the
2466 % examining board, and Annex D shows nothing below the signature lines. The date
2467 % that belongs on this page is the date of APPROVAL, printed above.
2468 % \frontpage@bottomtext stays defined for anyone calling it directly.
2469 \vspace*{\fill}}

2470 \newcommand\coppe@hypersetup{%
2471 \begingroup
2472 % changes to \toks@ and \count@ are kept local;
2473 % it's not necessary for them, but it is usually the case
2474 % for \count1, because the first ten counters are written
2475 % to the DVI file, thus you got lucky because of PDF output
2476 \toks@={}% in this special case not necessary
2477 \count@=0 %
2478 \@whilenum\count@<\value{keywords}\do{%
2479     % * a keyword separator is not necessary,
2480     %   if there is just one keyword
2481     % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
2482     %   at least once, to get rid of the loop depended \count@
2483     \ifcase\count@ % same as \ifnum0=\count@
2484         \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
2485     \else
2486         \toks@=\expandafter\expandafter\expandafter{%
2487             \expandafter\the\expandafter\toks@

```

```

2488         \expandafter;\expandafter\space
2489         \csname CoppeKeyword:\the\count@\endcsname
2490     }%
2491     \fi
2492     \advance\count@ by 1 %
2493 }%
2494 \edef\x{\endgroup
2495     \noexpand\hypersetup{%
2496         pdfkeywords={\the\toks@}%
2497     }%
2498 }%
2499 \x
2500 \hypersetup{%
2501     pdfauthor={\@authname\ \@authsur},
2502     pdftitle={\local@title},
2503     pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
2504     pdfcreator={LaTeX with CoppeTeX toolkit},
2505     breaklinks={true},
2506     raiselinks={true},
2507     pageanchor={true},
2508 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.,” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\isundefined` (cf. the T_EX FAQ).

```

2509 \newcommand\makecatalog{%
2510     \vspace*{\fill}
2511     \begin{center}
2512         \setlength{\fboxsep}{5mm}
2513         \framebox[120mm][c]{\makebox[5mm][c]{}%
2514             \begin{minipage}[c]{105mm}
2515                 \setlength{\parindent}{5mm}
2516                 \noindent\sloppy\nohyphens\@authsur,
2517                 \nohyphens\@authname\par
2518                 \nohyphens{%
2519                     \coppe@selecttitle/\@authname\ \@authsur. -- \local@cityname:
2520                     UFRJ/COPPE, \coppe@docyear.}\par
2521                 \pageref{front:pageno},
2522                 \pageref{LastPage}
2523                 p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.\par
2524                 % There is an issue here. When the last entry must be split between lines,
2525                 % the spacing between it and the next paragraph becomes smaller.
2526                 % Should we manually introduce a fixed space? But how could we know that
2527                 % a name was split? Is this happening yet?
2528                 \nohyphens{%
2529                     \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2530                         \local@advisorstring: &
2531                         \count1=0
2532                         \toks@={}
2533                         \@whilenum \count1<\@advisor \do{%
2534                             \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2535                                 \expandafter\endcsname\expandafter\space%

```



```

2537         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2538     \else
2539         \toks@=\expandafter\expandafter\expandafter{%
2540             \expandafter\the\expandafter\toks@
2541             \expandafter&\expandafter\space
2542             \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2543             \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2544         }%
2545     \fi
2546     \advance\count1 by 1}
2547     \the\toks@
2548 \end{tabularx}}\par
2549 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
2550 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
2551 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
2552 \count1=0
2553 \count2=1
2554 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
2555     \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
2556     \advance\count1 by 1
2557     \advance\count2 by 1}
2558 I. \csname CoppeAdvisorSurname:0\endcsname,%
2559 \ \csname CoppeAdvisorName:0\endcsname%
2560 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
2561 II. \local@universityname, COPPE, Programa de \local@deptname.
2562 III. T{\' i}tulo.}
2563 \end{minipage}}
2564 \end{center}
2565 \vspace*{\fill}}

```

`\fichacatalografica` The folha adicional required from August 2026 (SiBI manual, 9th rev. ed., `\makefolhaadicional` 3.1.2.1.2 and Annex H): the CAPES collection fields plus the ficha catalografica, sitting immediately after the folha de rosto and counted by nobody. `\fichacatalografica` takes the PDF produced by the SiBI generator at <http://fichacatalografica.sibi.ufrj.br/>, optionally with a page number.

The sheet is institutional, so it is always in Portuguese whatever the main language of the work – and deliberately without `\selectlanguage`, since the brazilian babel language need not be loaded in, say, a Spanish-main document.

```

2566 \newcommand\coppe@fichafila{}
2567 \newcommand\coppe@fichapage{1}
2568 \newcommand\fichacatalografica[2][1]{%
2569     \gdef\coppe@fichapage{#1}\gdef\coppe@fichafila{#2}}
2570 \newcommand\coppe@marcatipo[1]{%
2571     \ifthenelse{\equal{\coppe@tipoproducao}{#1}}{$\times$}{\phantom{$\times$}}}
2572 \newcommand\coppe@marcavinc[1]{%
2573     \ifthenelse{\equal{\coppe@projetovinc}{#1}}{$\times$}{\phantom{$\times$}}}
2574 \newcommand\coppe@campo[1]{%
2575     \ifthenelse{\equal{#1}}{\hrulefill}{#1\enspace\hrulefill}}
2576 \newcommand\coppe@checatipo{%
2577     \ifthenelse{\equal{\coppe@tipoproducao}{}}{}}{%
2578     \ifthenelse{\equal{\coppe@tipoproducao}{bibliografica}}\OR
2579         \equal{\coppe@tipoproducao}{artistica}\OR
2580         \equal{\coppe@tipoproducao}{tecnologica}\OR

```

```

2581 \equal{\coppe@tipoproducao}{tecnica}}{ }{%
2582 \ClassWarning{coppe}{Unknown \string\tipoproducao{ } value
2583 '\coppe@tipoproducao'. Annex H of the SiBI manual allows only
2584 bibliografica, artistica, tecnologica or tecnica; no box will be
2585 ticked}}}%
2586 \newcommand\makefolhaadicional{%
2587 \clearpage
2588 \thispagestyle{empty}%
2589 \coppe@checatipo
2590 {\singlespacing
2591 \begin{center}
2592 {\large\bfseries Informa\c c\~oes Coleta CAPES}
2593 \end{center}
2594 \vspace{7mm}
2595 \noindent Tipo de produ\c c\~ao intelectual:\par
2596 \vspace{1mm}
2597 \hspace*{10mm}(\coppe@marcatipo{bibliografica})\enspace Bibliogr\'afica\par
2598 \hspace*{10mm}(\coppe@marcatipo{artistica})\enspace Art\'istica\par
2599 \hspace*{10mm}(\coppe@marcatipo{tecnologica})\enspace Tecnol\'ogica\par
2600 \hspace*{10mm}(\coppe@marcatipo{tecnica})\enspace T\'ecnica\par
2601 \vspace{5mm}
2602 \noindent Projeto de pesquisa vinculado \'a produ\c c\~ao intelectual?\par
2603 \vspace{1mm}
2604 \hspace*{10mm}(\coppe@marcavinc{sim})\enspace Sim\par
2605 \hspace*{10mm}(\coppe@marcavinc{nao})\enspace N\~ao\par
2606 \vspace{5mm}
2607 \noindent Nome do projeto de pesquisa:\par
2608 \vspace{1mm}
2609 \noindent\coppe@campo{\coppe@nomeprojeto}\par
2610 \vspace{5mm}
2611 \noindent \'Area de concentra\c c\~ao da produ\c c\~ao intelectual:\par
2612 \vspace{1mm}
2613 \noindent\coppe@campo{\coppe@areaconc}\par
2614 \vspace{5mm}
2615 \noindent Ag\~encia(s) de fomento financiadora(s) da produ\c c\~ao
2616 intelectual:\par
2617 \vspace{1mm}
2618 \ifthenelse{\value{coppeagencias}=0}{%
2619 \noindent\hrulefill\par}{%
2620 \count1=0
2621 \@whilenum\count1<\value{coppeagencias}\do{%
2622 \noindent\hspace*{10mm}\csname CoppeAgencia:\the\count1\endcsname
2623 \ (\csname CoppeAgenciaSigla:\the\count1\endcsname)\par
2624 \advance\count1 by 1}}%
2625 \vspace{9mm}
2626 \begin{center}
2627 {\large\bfseries Ficha catalogr\'afica}\par
2628 \vspace{4mm}
2629 \ifthenelse{\equal{\coppe@fichafile}{ } }{%
2630 \if@copperascunho
2631 \makecatalog
2632 \else
2633 \framebox[125mm][c]{\parbox[c][62mm][c]{115mm}{\centering\footnotesize
2634 Espa\c co reservado \'a ficha catalogr\'afica.\\[1ex]

```

```

2635         Gere-a em \texttt{fichacatalografica.sibi.ufrj.br} e inclua o
2636         arquivo com \texttt{\string\fichacatalografica\{arquivo.pdf\}},
2637         ou solicite-a \‘a biblioteca do seu Programa.}\}%
2638     \fi}{%
2639     \includegraphics [page=\coppe@fichapage,width=125mm]{\coppe@fichafile}}%
2640 \end{center}
2641 \par}}

```

\dedication

```

2642 \newcommand\dedication[1]{
2643   \gdef\@dedic{#1}
2644   \cleardoublepage
2645   \vspace*{\fill}
2646   \begin{flushright}
2647     \begin{minipage}{60mm}
2648       \raggedleft \it \normalsize \@dedic
2649     \end{minipage}
2650   \end{flushright}}

```

\coppe@coadvisorrows The coadvisor block of the abstract pages, in the shape the folha de rosto already uses. It prints nothing at all unless the `coorientador` class option is given AND the document actually declared a `\coadvisor`, so the default output is byte for byte what it was. The argument is the label macro, which differs per page: the main-language pages pass `\local@coadvisorstring` and the foreign one passes `\foreign@coadvisorstring`.

```

2651 \newcommand\coppe@coadvisorrows[1]{%
2652   \if@coppecoadvresumo
2653     \ifnum\@coadvisor>0
2654       \par\vspace*{2mm}%
2655       \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2656         #1: &
2657         \count1=0
2658         \toks@={}
2659         \@whilenum \count1<\@coadvisor \do{%
2660           \ifcase\count1
2661             \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2662               \expandafter\endcsname\expandafter\space%
2663               \csname CoppeCoadvisorSurname:\the\count1\endcsname\}}
2664           \else
2665             \toks@=\expandafter\expandafter\expandafter{%
2666               \expandafter\the\expandafter\toks@%
2667               \expandafter\&\expandafter\space%
2668               \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2669               \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\}
2670             }%
2671           \fi
2672           \advance\count1 by 1}
2673         \the\toks@
2674       \end{tabularx}%
2675     \fi
2676   \fi}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

2677 \newenvironment{abstract}{%
2678 \cleardoublepage
2679 \thispagestyle{plain}
2680 \abstract@toptext\par
2681 \vspace*{8.6mm}
2682 \begin{center}
2683 \sloppy\nohyphens{\MakeUppercase\local@title}\par
2684 \vspace*{13.2mm}
2685 \@authname\ \@authsur \par
2686 \vspace*{7mm}
2687 \local@monthname/\coppe@docyear
2688 \end{center}\par
2689 \vspace*{1.5cm}
2690 \noindent%
2691 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}
2692 \local@advisorstring: &
2693 \count1=0
2694 \toks@={}
2695 \@whilenum \count1<\@advisor \do{%
2696 \ifcase\count1 % same as \ifnum0=\count1
2697 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2698 \expandafter\endcsname\expandafter\space%
2699 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2700 \else
2701 \toks@=\expandafter\expandafter\expandafter{%
2702 \expandafter\the\expandafter\toks@
2703 \expandafter&\expandafter\space
2704 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2705 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2706 }%
2707 \fi
2708 \advance\count1 by 1}
2709 \the\toks@
2710 \end{tabularx}%
2711 \coppe@coadvisorrows{\local@coadvisorstring}\par
2712 \vspace*{2mm}
2713 \noindent\local@deptstring: \local@deptname\par
2714 \vspace*{7mm}}{%
2715 \coppe@printkeywords{\coppemainstring{keywordsname}}%
2716 {CoppeKeyword:}{\value{keywords}}}%
2717 \vspace*{\fill}}

```

foreignabstract (env.)

```

2718 \newenvironment{foreignabstract}{%
2719 \cleardoublepage
2720 \thispagestyle{plain}
2721 \begin{otherlanguage}{\coppe@foreignlang}
2722 \foreignabstract@toptext\par
2723 \vspace*{8.6mm}
2724 \begin{center}
2725 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
2726 \vspace*{13.2mm}
2727 \@authname\ \@authsur \par
2728 \vspace*{7mm}

```

```

2729 \foreign@monthname/\coppe@docyear
2730 \end{center}\par
2731 \vspace*{1.5cm}
2732 \noindent%
2733 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2734 \foreign@advisorstring: &
2735 \count1=0
2736 \toks@={}
2737 \@whilenum \count1<\@advisor \do{%
2738 \ifcase\count1 % same as \ifnum0=\count1
2739 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2740 \expandafter\endcsname\expandafter\space%
2741 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2742 \else
2743 \toks@=\expandafter\expandafter\expandafter{%
2744 \expandafter\the\expandafter\toks@
2745 \expandafter&\expandafter\space
2746 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2747 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2748 }%
2749 \fi
2750 \advance\count1 by 1}
2751 \the\toks@
2752 \end{tabularx}%
2753 \coppe@coadvisorrows{\foreign@coadvisorstring}\par
2754 \vspace*{2mm}
2755 \noindent\foreign@deptstring: \foreign@deptname\par
2756 \vspace*{7mm}}}%
2757 \ifnum\value{foreignkeywords}>0
2758 \coppe@printkeywords{\coppeforeignstring{keywordsname}}}%
2759 {CoppeForeignKeyword:}{\value{foreignkeywords}}}%
2760 \else
2761 \coppe@printkeywords{\coppeforeignstring{keywordsname}}}%
2762 {CoppeKeyword:}{\value{keywords}}}%
2763 \fi
2764 \end{otherlanguage}
2765 \vspace*{\fill}
2766 \global\let\@author\@empty
2767 \global\let\@date\@empty
2768 \global\let\foreign@title\@empty
2769 \global\let\foreign@title\relax
2770 \global\let\local@title\@empty
2771 \global\let\local@title\relax
2772 \global\let\author\relax
2773 \global\let\author\relax
2774 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor

english; harmless otherwise (the document just won't include it).

```

2775 \newenvironment{brazilianabstract}{%
2776   \cleardoublepage
2777   \thispagestyle{plain}
2778   \begin{otherlanguage}{brazilian}
2779   \abstract@toptext\par
2780   \vspace*{8.6mm}
2781   \begin{center}
2782     \sloppy\nohyphens{\MakeUppercase\local@title}\par
2783     \vspace*{13.2mm}
2784     \@authname\ \@authsur\n \par
2785     \vspace*{7mm}
2786     \local@monthname/\coppe@docyear
2787   \end{center}\par
2788   \vspace*{1.5cm}
2789   \noindent%
2790   \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}
2791     \local@advisorstring: &
2792     \count1=0
2793     \toks@={}
2794     \@whilenum \count1<\@advisor \do{%
2795       \ifcase\count1
2796         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2797           \expandafter\endcsname\expandafter\space%
2798           \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2799       \else
2800         \toks@=\expandafter\expandafter\expandafter{%
2801           \expandafter\the\expandafter\toks@
2802           \expandafter&\expandafter\space
2803           \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2804           \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2805         }%
2806       \fi
2807       \advance\count1 by 1}
2808     \the\toks@
2809   \end{tabularx}%
2810   \coppe@coadvisorrows{\local@coadvisorstring}\par
2811   \vspace*{2mm}
2812   \noindent\local@deptstring: \local@deptname\par
2813   \vspace*{7mm}}{%
2814   % \coppestring takes ONE argument and resolves against the current language;
2815   % this runs before \end{otherlanguage}, so \language is already brazilian.
2816   \ifnum\value{braziliankeywords}>0
2817     \coppe@printkeywords{\coppestring{keywordsname}}%
2818     {CoppeBrazilianKeyword:}{\value{braziliankeywords}}%
2819   \else
2820     \coppe@printkeywords{\coppestring{keywordsname}}%
2821     {CoppeKeyword:}{\value{keywords}}%
2822   \fi
2823   \end{otherlanguage}
2824   \vspace*{\fill}}

```

\listoffigures

```

2825 \renewcommand\listoffigures{%

```

```

2826 \coppe@hasLof
2827 \if@twocolumn
2828 \@restonecoltrue\onecolumn
2829 \else
2830 \@restonecolfalse
2831 \fi
2832 \chapter*{\listfigurename}%
2833 \coppe@listtoc{\listfigurename}%
2834 \mkboth{\MakeUppercase\listfigurename}%
2835 {\MakeUppercase\listfigurename}%
2836 \@starttoc{lof}%
2837 \if@restonecol\twocolumn\fi
2838 }

```

\listoftables

```

2839 \renewcommand\listoftables{%
2840 \if@twocolumn
2841 \@restonecoltrue\onecolumn
2842 \else
2843 \@restonecolfalse
2844 \fi
2845 \chapter*{\listtablename}%
2846 \coppe@listtoc{\listtablename}%
2847 \mkboth{%
2848 \MakeUppercase\listtablename}%
2849 {\MakeUppercase\listtablename}%
2850 \@starttoc{lot}%
2851 \if@restonecol\twocolumn\fi
2852 }

```

\printlosymbols

```

2853 \newcommand\printlosymbols{%
2854 \renewcommand\glossaryname{\listsymbolname}%
2855 \begingroup\@coppeglosspretrue
2856 \@input@{\jobname.los}\endgroup}

```

\makelosymbols

```

2857 \def\makelosymbols{%
2858 \newwrite\@losfile
2859 \immediate\openout\@losfile=\jobname.syx
2860 \newcommand\symb1[3][\@bsphack\begingroup
2861 \ifstrepty{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
2862 \@sanitize%
2863 \@wrlos{\@tempsymb1}{##2}{##3}\typeout%
2864 {Writing index of symbols file \jobname.syx}%
2865 \let\makelosymbols\@empty%
2866 }%
2867 \@onlypreamble\makelosymbols

2868 \AtBeginDocument{%
2869 \@ifpackageloaded{hyperref}{%
2870 \newcommand\@wrlos[3]{%
2871 \protected@write\@losfile{%
2872 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%

```

```

2873 \endgroup%
2874 \@esphack}}{%
2875 \newcommand\@wrls[3]{%
2876 \protected@write\@losfile{%
2877 {\string\indexentry{#1[#2] #3}{\thepage}}%
2878 \endgroup%
2879 \@esphack}}}%

```

\printloabbreviations

```

2880 \newcommand\printloabbreviations{%
2881 \renewcommand\glossaryname{\listabbreviationname}%
2882 \begingroup\@coppeglosspretrue
2883 \@input{\jobname.lab}\endgroup}

```

\makeloabbreviations

```

2884 \def\makeloabbreviations{%
2885 \newwrite\@labfile
2886 \immediate\openout\@labfile=\jobname.abx
2887 \newcommand\abbrev[3][\@bsphack\begingroup
2888 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
2889 \@sanitize
2890 \@wrlab{\@tempsymb1}{##2}{##3}\typeout
2891 {Writing index of abbreviations file \jobname.abx}%
2892 \let\makeloabbreviations\@empty
2893 }
2894 \@onlypreamble\makeloabbreviations

2895 \AtBeginDocument{%
2896 \@ifpackageloaded{hyperref}{%
2897 \newcommand\@wrlab[3]{%
2898 \protected@write\@labfile{%
2899 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2900 \endgroup%
2901 \@esphack}}{%
2902 \newcommand\@wrlab[3]{%
2903 \protected@write\@labfile{%
2904 {\string\indexentry{#1[#2] #3}{\thepage}}%
2905 \endgroup%
2906 \@esphack}}}%

```

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
 \sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full form. The first-use flag is set globally so it survives groups (floats, footnotes).

```

2907 \newcommand\newsigla[3]{%
2908 \@namedef{coppe@sigla@s@#1}{#2}%
2909 \@namedef{coppe@sigla@l@#1}{#3}%
2910 \newcommand\coppe@sigla@reg[1]{%
2911 \@ifundefined{abbrev}{%
2912 {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
2913 \newcommand\coppe@sigla@full[1]{%
2914 \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
2915 \newcommand\coppe@sigla@undef[1]{%

```



```

2916 \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
2917 \textbf{??#1??}}
2918 \newcommand\coppe@sigla@first[1]{%
2919 \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
2920 \newcommand\coppe@sigla@norm[1]{%
2921 \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2922 {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
2923 {\@nameuse{coppe@sigla@s@#1}}}}
2924 \newcommand\coppe@sigla@star[1]{%
2925 \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2926 {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}}
2927 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

2928 %%% \AtBeginDocument{%
2929 %%% \ifpackageloaded{hyperref}{%
2930 %%% \def\@wrlab#1#2{%
2931 %%% \protected@write\@labfile{%
2932 %%% {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
2933 %%% \endgroup
2934 %%% \@esphack}}{%
2935 %%% \def\@wrlab#1#2{%
2936 %%% \protected@write\@labfile{%
2937 %%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
2938 %%% \endgroup
2939 %%% \@esphack}}}

2940 % Catalog-label writers. Two language-related traps must be avoided here:
2941 % 1. The label NAMES contain ':' and '.', which babel makes active in
2942 % some languages (e.g. Brazilian's "shorthand" set). If the write
2943 % argument is tokenised while one of those is active, pdfTeX aborts
2944 % with "<inserted text> }\endwrite" because the shorthand expansion
2945 % injects unbalanced braces. \detokenize emits its argument as
2946 % catcode-12 characters regardless of the current activation, so the
2947 % label NAMES survive whatever language is current at writer time.
2948 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
2949 % produce small-caps roman numerals via \es@scroman, whose expansion
2950 % contains an \uppercase group that is unbalanced inside \write. The
2951 % same crash signature appears. The fix is to bypass LaTeX's
2952 % \roman/\arabic counter wrappers entirely and use the raw e-TeX
2953 % primitives \romannumeral and \number directly on the \c@page count
2954 % register --- both are fully expandable, language-agnostic, and
2955 % always produce plain ASCII digits or lowercase roman letters.
2956 % \Roman{page} is left in place because babel-spanish only redefines
2957 % \@Roman under the (non-default) ucroman option.
2958 \AtBeginDocument{%
2959 \ltx@ifpackageloaded{hyperref}{%
2960 \def\coppe@bibEnd{%
2961 \immediate\write\@auxout{%
2962 \string\newlabel{\detokenize{bib:end}}%
2963 {{}\number\c@page}{\detokenize{page.}\number\c@page}}}}%
2964 \def\coppe@bibBegin{%
2965 \immediate\write\@auxout{%
2966 \string\newlabel{\detokenize{bib:begin}}%
2967 {{}\number\c@page}{\detokenize{page.}\number\c@page}}}}%
2968 \def\coppe@mainBegin{%

```

```

2969 \immediate\write\@auxout{%
2970 \string\newlabel{\detokenize{front:pageno}}}%
2971 {}{\Roman{page}}{}{\detokenize{page.}\romannumeral\c@page}{}}}%
2972 \def\coppe@hasLof{%
2973 \immediate\write\@auxout{%
2974 \string\newlabel{\detokenize{cat:lofflag}}}%
2975 {}{\detokenize{:~il.}}{}{\detokenize{page.}\romannumeral\c@page}{}}}%
2976 }{%
2977 \def\coppe@bibEnd{%
2978 \immediate\write\@auxout{%
2979 \string\newlabel{\detokenize{bib:end}}{}{\number\c@page}{}}}%
2980 \def\coppe@bibBegin{%
2981 \immediate\write\@auxout{%
2982 \string\newlabel{\detokenize{bib:begin}}{}{\number\c@page}{}}}%
2983 \def\coppe@mainBegin{%
2984 \immediate\write\@auxout{%
2985 \string\newlabel{\detokenize{front:pageno}}{}{\Roman{page}}{}}}%
2986 \def\coppe@hasLof{%
2987 \immediate\write\@auxout{%
2988 \string\newlabel{\detokenize{cat:lofflag}}{}{\detokenize{:~il.}}{}}}%
2989 }%
2990 }
2991 % The reference list is now produced by biblatex's \printbibliography (see the
2992 % bibliography setup after babel, above). The former thebibliography
2993 % redefinition and its \bibindent are no longer used and have been removed; the
2994 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
2995 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

2996 \newlength{\recuolongquote}
2997 \setlength{\recuolongquote}{4cm}
2998
2999 \newenvironment*{longquote}[1][default]{%
3000 \begin{list}{}{%
3001 \setlength{\leftmargin}{\recuolongquote}%
3002 \setlength{\rightmargin}{0pt}%
3003 \setlength{\itemindent}{0pt}%
3004 \setlength{\listparindent}{0pt}%
3005 \setlength{\parsep}{0pt}%
3006 \setlength{\topsep}{\baselineskip}%
3007 }%
3008 \item\relax
3009 \footnotesize
3010 \singlespacing
3011 \ifthenelse{\not\equal{#1}{default}}{%
3012 {\itshape\selectlanguage{#1}}%
3013 }%
3014 \ignorespaces
3015 }{%
3016 \end{list}%
3017 \ignorespacesafterend
3018 }
3019 \newenvironment{theglossary}{%

```

```

3020 \if@twocolumn%
3021   \@restonecoltrue\onecolumn%
3022 \else%
3023   \@restonecolfalse%
3024 \fi%
3025 \@mkboth{\MakeUppercase\glossaryname}%
3026 {\MakeUppercase\glossaryname}%
3027 \chapter*{\glossaryname}%
3028 \if@coppeglosspre
3029   \coppe@listtoc{\glossaryname}%
3030 \else
3031   \addcontentsline{toc}{chapter}{\glossaryname}%
3032 \fi
3033 \list{}
3034 {\setlength{\listparindent}{0in}%
3035  \setlength{\labelwidth}{1.0in}%
3036  \setlength{\leftmargin}{1.5in}%
3037  \setlength{\labelsep}{0.5in}%
3038  \setlength{\itemindent}{0in}}%
3039 \sloppy}%
3040 {\if@restonecol\twocolumn\fi%
3041 \endlist}
3042 %
3043 \renewenvironment{theindex}{%
3044   \if@twocolumn
3045     \@restonecolfalse
3046   \else
3047     \@restonecoltrue
3048   \fi
3049   \twocolumn[\@makeschapterhead{\indexname}]%
3050   \@mkboth{\MakeUppercase\indexname}%
3051   {\MakeUppercase\indexname}%
3052   \thispagestyle{plain}\parindent\z@
3053   \addcontentsline{toc}{chapter}{\indexname}
3054   \parskip\z@ \@plus .3\p@\relax
3055   \columnseprule \z@
3056   \columnsep 35\p@
3057   \let\item\@idxitem}
3058   {\if@restonecol\onecolumn\else\clearpage\fi}
3059 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
3060 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
3061 \newcommand\glossaryname{\coppemainstring{glossary}}
3062 %
3063 \newcommand\local@advisorstring{Orientador}
3064 \newcommand\foreign@advisorstring{Advisor}
3065 % Fed by the string dispatcher instead of a hard-coded Portuguese literal.
3066 % Expansion is lazy (it happens inside \makefrontpage), so \coppe@mainlang is
3067 % already set by the time this is used.
3068 \ifthenelse{\boolean{maledoc}}{%
3069   \newcommand\local@approvedname{\coppemainstring{approvedmale}}%
3070 }{%
3071   \newcommand\local@approvedname{\coppemainstring{approvedfemale}}%
3072 }
3073 \ifthenelse{\boolean{maledoc}}{%

```

```

3074 \newcommand\local@approvedonname{\coppemainstring{approvedonmale}}}%
3075 }{%
3076 \newcommand\local@approvedonname{\coppemainstring{approvedonfemale}}}%
3077 }
3078 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
3079 \newcommand\local@deptstring{Programa}
3080 \newcommand\foreign@deptstring{Department}
3081 \newcommand\local@cityname{Rio de Janeiro}
3082 \newcommand\local@statename{RJ}
3083 \newcommand\local@countryname{Brasil}
3084 %
3085 % 3.1.2.1.1(e) of the 2026 manual requires the natureza, the objetivo, the
3086 % name of the institution AND the area de concentracao in this block. The area
3087 % was missing; it is appended as its own sentence when \areaconcentracao was
3088 % given, and simply omitted otherwise.
3089 \newcommand\frontcover@maintext{
3090 \sloppy\nohyphens{\local@doctype\ de \@degreename\
3091 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3092 ao Programa de P{\ ' o}s-gradua{\c c}{\~ a}o em \local@deptname,
3093 COPPE, da \local@universityname, como parte dos requisitos
3094 necess{\ ' a}rios {\ ' a} obten{\c c}{\~ a}o do t{\ ' i}tulo de
3095 \local@degname\ em \local@deptname.}%
3096 \ifthenelse{\equal{\coppe@areaconc}{}}{ }{%
3097 \par\vspace{1ex}}%
3098 \sloppy\nohyphens{\ ' Area de concentra{\c c}{\~ a}o: \coppe@areaconc.}}%
3099 }
3100 %
3101 \newcommand\frontpage@maintext{
3102 \noindent {\MakeUppercase\local@doctype}
3103 \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
3104 \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
3105 DE P{\ ' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
3106 UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
3107 NECESS{\ ' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
3108 {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
3109 {\MakeUppercase\local@deptname.\par}}%
3110 }
3111 %
3112 \newcommand\frontpage@bottomtext{%
3113 \begin{center}
3114 {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
3115 {\MakeUppercase\local@monthname\ DE \coppe@docyear}
3116 \end{center}}%
3117 }
3118 %
3119 \newcommand\abstract@toptext{%
3120 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
3121 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3122 \sloppy\nohyphens{\ ' a} COPPE/UFRJ como parte dos requisitos
3123 necess{\ ' a}rios para a obten{\c c}{\~ a}o do grau de
3124 \local@degname\ em Ci{\^e}ncias (\@degree)}
3125 }
3126 \newcommand\foreignabstract@toptext{%
3127 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to

```

```

3128 COPPE/UFRJ as a partial fulfillment of the requirements for the
3129 degree of \foreign@degname\ of Science (\@degree)}
3130 }
3131 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, `list-of-*` commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquadros`).
- **biblatex** – must come after **hyperref** so its citation links and back-references hook into **hyperref** correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

3132 \RequirePackage{setspace}
3133 % Apply the default line spacing now that setspace is available. The flag was
3134 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
3135 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
3136 % pdfx MUST precede hyperref -- it loads hyperref itself, with the options
3137 % PDF/A needs, and cannot fix up an already-loaded one. Level a-2b rather than
3138 % a-1b (the "ISO 19005-1" of Annex I): a-1b forbids transparency, which
3139 % tcolorbox and several graphics packages produce, so a-1b would fail on
3140 % documents that are otherwise perfectly fine. a-2b is ISO 19005-2, accepted by
3141 % Pantheon and by the SiBI's own instructions, which also list PDF/A-2b.
3142 \if@coppepdfa
3143   \RequirePackage[a-2b]{pdfx}
3144 % pdfTeX stamps /PTEX.Fullbanner into the document information dictionary.
3145 % It is not one of the keys ISO 19005-2 6.6.2.3.2 maps to an XMP property, and
3146 % no extension schema declares it, so it is a DocInfo entry with no XMP
3147 % counterpart -- a finding a PDF/A checker has every reason to raise, and one
3148 % that costs nothing to remove. -1 sets every suppression bit.
3149 %
3150 % The two engines spell this differently, and not just in the name. pdfTeX has
3151 % \pdfsuppressptexinfo, whose bits only ever govern PTEX.* entries, so -1 (all
3152 % of them) is safe. LuaTeX moved the pdfTeX integer parameters behind
3153 % \pdfvariable in 0.85 AND renamed this one to suppressoptionalinfo, whose bits
3154 % reach further: besides the four PTEX.* entries (1, 2, 4, 8) it can also
3155 % suppress Creator, the dates, Producer, Trapped and -- the dangerous one --
3156 % the trailer /ID, which PDF/A requires. So -1 is exactly what must NOT be
3157 % written there; 15 is the four PTEX.* bits and nothing else. Checked on a real
3158 % LuaTeX run: /ID survives.
3159 %
3160 % Getting the name wrong is not a quiet failure. \pdfvariable with an unknown
3161 % variable name makes LuaTeX typeset the name, and the document stops with

```

```

3162 % "Missing \begin{document}" -- which is how the first attempt at this was
3163 % caught.
3164 \ifdefined\pdfsuppressptexinfo
3165 \pdfsuppressptexinfo=-1
3166 \else
3167 \ifdefined\pdfvariable
3168 \pdfvariable suppressoptionalinfo=15
3169 \fi
3170 \fi
3171 % pdfx takes the XMP metadata from \jobname.xmpdata, which it reads when it is
3172 % LOADED -- before the preamble has run, so before \title, \author and
3173 % \keyword exist. The file is therefore WRITTEN by the class and picked up on
3174 % the NEXT run. The build already runs pdflatex three times for cross-
3175 % references, so the deposited PDF always carries current metadata; only a
3176 % first-ever compilation of a brand-new file emits pdfx's "no \jobname.xmpdata"
3177 % warning and lacks dc:title.
3178 %
3179 % The write happens at the END of the document, not at \begin{document}. This
3180 % class has always documented -- and every example follows it -- that \title,
3181 % \author, \department and \keyword go AFTER \begin{document}, next to
3182 % \maketitle. A write at \begin{document} therefore ran before any of them
3183 % existed and died with "Undefined control sequence" on \@authname, so the
3184 % pdfa option was unusable in exactly the style the manual teaches. It only
3185 % ever worked for a document that set its metadata in the preamble. At the end
3186 % everything is set whichever style the author chose, and \immediate still puts
3187 % the file on disk in time for the next run.
3188 %
3189 % The hook is enddocument/afteraux, and the stream is \@mainaux rather than one
3190 % of our own. TeX has sixteen output streams and example.tex was already using
3191 % all of them (toc, lof, lot, loa, lol, loq, los, idx, glo, aux, bcf, ...), so
3192 % \newwrite here failed with "No room for a new \write" on precisely the
3193 % document that most needs validating. enddocument/afteraux fires after LaTeX
3194 % has closed the .aux, which frees that stream for the one \immediate\openout
3195 % we need and costs the document nothing. Kernels older than 2020-10-01 have no
3196 % such hook; there the class falls back to \AtEndDocument with a stream of its
3197 % own, which is the pre-existing behaviour.
3198 %
3199 % Every field is guarded, and the guards are not cosmetic. \coppe@selecttitle
3200 % is \@nameuse{coppe@title<lang>}, and \csname on a name that does not exist
3201 % yet MAKES it \relax: unexpandable, so \write puts the control sequence NAME
3202 % into the file instead of the title. The .xmpdata then carried the literal
3203 % \coppe@title@brazilian, pdfx read it back on the following run, and the
3204 % document stopped compiling at all -- a single bad run bricked the file until
3205 % someone deleted the .xmpdata by hand. A field that is not set is now simply
3206 % omitted, which pdfx handles, instead of poisoning the next run.
3207 %
3208 % The writes must be \immediate. \protected@write defers to the shipout, and
3209 % the stream is closed here and now, so deferred writes would land nowhere --
3210 % which is exactly what happened first time round: a 0-byte .xmpdata.
3211 % \let\protect\string inside the group gives the protection that
3212 % \protected@write would have given: accents and other robust commands are
3213 % written literally instead of being expanded. pdfx reads the .xmpdata as
3214 % LaTeX, so \'a arrives there intact.
3215 \ifdefined\AddToHook

```

```

3216 \let\coppe@xmpout\@mainaux
3217 \def\coppe@xmphook{\AddToHook{enddocument/afteraux}}%
3218 \else
3219 \newwrite\coppe@xmpout
3220 \let\coppe@xmphook\AtEndDocument
3221 \fi
3222 \coppe@xmphook{%
3223 \gdef\coppe@xmpkeys{%
3224 \count@=0
3225 \@whilenum\count@<\value{keywords}\do{%
3226 \ifnum\count@>0
3227 \g@addto@macro\coppe@xmpkeys{\string\sep\space}%
3228 \fi
3229 \protected@edef\coppe@tmp{\csname CoppeKeyword:\the\count@\endcsname}%
3230 \expandafter\g@addto@macro\expandafter\coppe@xmpkeys
3231 \expandafter{\coppe@tmp}%
3232 \advance\count@ by 1}%
3233 \immediate\openout\coppe@xmpout=\jobname.xmpdata
3234 \begingroup
3235 \let\protect\string
3236 \@ifundefined{coppe@title@\coppe@mainlang}{}%{
3237 \immediate\write\coppe@xmpout{\string\Title{\coppe@selecttitle}}}%
3238 \@ifundefined{@authname}{}%{
3239 \immediate\write\coppe@xmpout{
3240 \string\Author{\@authname\space\@authsur}}}%
3241 \immediate\write\coppe@xmpout{\string\Keywords{\coppe@xmpkeys}}%
3242 \ifthenelse{\equal{\coppe@deptcode}{}}{ }{%
3243 \immediate\write\coppe@xmpout{\string\Subject{\local@doctype\space de
3244 \@degreename. Programa de \meta@deptname\space(\coppe@deptcode),
3245 COPPE/UFRJ}}}%
3246 \immediate\write\coppe@xmpout{\string\Publisher{\local@universityname}}%
3247 \@ifundefined{coppe@docyear}{}%{
3248 \immediate\write\coppe@xmpout{\string\Date{\coppe@docyear}}}%
3249 \endgroup
3250 \immediate\closeout\coppe@xmpout}%
3251 \fi
3252 \RequirePackage{hyperref}
3253 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
3254 % former natbib+bibtex setup. The numbers class option selects the numeric
3255 % citation style; otherwise the author-date default is used. natbib=true keeps
3256 % \citet/\citep available for users migrating from natbib.
3257 \if@coppenumeric
3258 \RequirePackage[backend=biber,database=coppe,%
3259 bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{bibtex}
3260 \else
3261 \RequirePackage[backend=biber,database=coppe,%
3262 bibstyle=coppe,citestyle=coppe,natbib=true]{bibtex}
3263 \fi
3264 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
3265 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
3266 % "Bibliografia" (babel's default \bibname). The title is rendered through
3267 % \coppestring so it follows the current main language in BOTH the chapter
3268 % heading and the .toc entry (bibtex's \bibstring proved unreliable across
3269 % the .toc write, and babel's \bibname caption override was equally fragile).

```

```

3270 \defbibheading{bibliography}{%
3271   \chapter*{\coppestring{references}}%
3272   \addcontentsline{toc}{chapter}{\coppestring{references}}}
3273 % The catalog page range (bib:begin/bib:end) was written by the old
3274 % thebibliography redefinition, which biblatex's \printbibliography does not
3275 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
3276 % begin label uses that hook and the end label is emitted by wrapping
3277 % \printbibliography to write it just after the reference list.
3278 \AtBeginBibliography{\coppe@bibBegin}
3279 \let\coppe@orig@printbibliography\printbibliography
3280 \renewcommand\printbibliography[1][]{%
3281   \coppe@orig@printbibliography[#1]%
3282   \coppe@bibEnd}
3283 \</class>
3284 \<*glossary>
3285 actual '='
3286 quote '!'
3287 level '>'
3288 %%%% delim_0 " , p. "
3289 delim_0 "\\dotfill "
3290 lethead_flag 0
3291 headings_flag 0
3292 preamble
3293 "\n\begin{theglossary}\n \\makeatletter"
3294 postamble
3295 "\n \\end{theglossary}\n"
3296 \</glossary>
3297 \<*class>
3298 \newcommand{\annex}{
3299   \renewcommand{\appendixname}{\coppestring{annex}}%
3300   \appendix
3301   \@coppeannextrue
3302 % \appendix has already re-issued the centred chapter format, but it baked in
3303 % the \appendixname of that moment. Re-issue it now that the name is "Anexo".
3304   \coppe@appendixheadings
3305 }
3306 \</class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

3307 \<*dbx>
3308 \ProvidesFile{coppe.dbx}[2026/05/28 v4.0 CoppeTeX ABNT datamodel]
3309 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
3310 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
3311 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
3312 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
3313 % performance) are already in biblatex's standard datamodel.
3314 \DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
3315 % Custom fields.
3316 \DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"
3317 \DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word

```



```

3318 \DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
3319 \DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
3320 \DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
3321 \DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
3322 \DeclareDatamodelFields[type=list,datatype=literal]{producer}
3323 \DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section
3324 \DeclareDatamodelFields[type=field,datatype=literal]{relator} % jurisprudence rapporteur
3325 \DeclareDatamodelEntryfields{course,coppedegree}
3326 \DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
3327 {artist,platform,director,producer,version}
3328 \DeclareDatamodelEntryfields[map]{scale}
3329 \DeclareDatamodelEntryfields[article]{caderno}
3330 \DeclareDatamodelEntryfields[jurisdiction]{relator}
3331 </dbx>
3332 <*bbx>
3333 \ProvidesFile{coppe.bbx}[2026/05/28 v4.0 CoppeTeX ABNT bibliography style]
3334 % The reference list is single-spaced (ABNT 2.4) via \singlespacing in
3335 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
3336 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
3337 % so load setspace here as well (idempotent).
3338 \RequirePackage{setspace}
3339 % ABNT reference list keeps the date at the end (imprenta), same for author-date
3340 % and numeric modes, so we derive from the standard style, not authoryear.
3341 \RequireBibliographyStyle{standard}
3342
3343 % Note: standard style never dashes repeated authors, which already matches the
3344 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
3345 \ExecuteBibliographyOptions{%
3346 maxbibnames=99, % ABNT 4.3.1.3: list all authors by default
3347 giveninits=false,
3348 uniquename=false,
3349 uniquelist=false,
3350 labeldateparts=true, % needed for author-date extradate (1999a/1999b)
3351 sorting=nyt,
3352 }
3353
3354 % Language strings (Disponível em / Acesso em / In, etc.)
3355 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
3356 \DeclareLanguageMapping{english}{english-coppe}
3357
3358 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
3359 \DeclareNameAlias{author}{family-given}
3360 \DeclareNameAlias{editor}{family-given}
3361 \DeclareNameAlias{translator}{family-given}
3362 \DeclareNameAlias{default}{family-given}
3363 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
3364 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
3365
3366 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
3367 % organizer prints the kind of responsibility in parentheses after the
3368 % name, abbreviated, lowercase and in the singular even for several
3369 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
3370 % standard style prints ", Org." (comma, no parentheses) when the editor
3371 % takes the author slot, so override editor+others to wrap the role string

```

```

3372 % in "...". The host position ("In: NAME (org.)") already parentheses
3373 % the role via coppe:bookeditor below. The role word itself is the
3374 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
3375 % packs (brazilian-coppe.lbx et al.).
3376 \renewbibmacro*{editor+others}{%
3377   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3378     {\printnames{editor}%
3379       \setunit*{\addspace}%
3380       \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
3381       \clearname{editor}}
3382   {}
3383 % Same parenthesised role for drivers that take the editor through the plain
3384 % editor macro instead of editor+others (e.g. @manual).
3385 \renewbibmacro*{editor}{%
3386   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
3387     {\printnames{editor}%
3388       \setunit*{\addspace}%
3389       \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3390       \clearname{editor}}
3391   {}
3392
3393 % Uppercase the surname (family + prefix) only inside the bibliography list;
3394 % citations keep normal caps (post-2020 rule).
3395 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
3396 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
3397 % field names; the month stays abbreviated via \mkbibmonth.
3398 \protected\def\coppe@abntdate#1#2#3{%
3399   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
3400   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
3401   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}%
3402 \AtBeginBibliography{%
3403   \singlespacing % entries single-spaced internally (2.4)
3404   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
3405   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
3406   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
3407   \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
3408   \let\mkbibdateshort\coppe@abntdate
3409 }
3410
3411 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
3412 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
3413 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
3414 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
3415 % article AND the next word are uppercased; otherwise just the first word. The
3416 % delimited macros split off words; \coppeUCkeep strips the trailing-space
3417 % sentinel so no spurious space is left before the following punctuation.
3418 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
3419 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
3420   \coppeUCifart{#1}
3421     {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
3422     {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
3423 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
3424   \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
3425 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}

```

```

3426 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
3427 \protected\def\coppeUCifart#1{%
3428   \begingroup
3429   \lowercase{\def\coppeUC@w{#1}}%
3430   \edef\coppeUC@w{\coppeUC@w}%
3431   \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,}%
3432   \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
3433   \endgroup}
3434 \DeclareFieldFormat{title}{%
3435   \ifboolexpr{ test {\ifnameundef{labelname}} }
3436     {#1}%
3437     {\mkbibbold{#1}}}
3438 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
3439 % format receives the RAW title text (the title format above only sees the
3440 % field-printing commands), so \coppeUCtitle can split off and uppercase the
3441 % first word here.
3442 \DeclareFieldFormat{titlecase}{%
3443   \ifboolexpr{ test {\ifnameundef{labelname}} }
3444     {\coppeUCtitle{#1}}%
3445     {#1}}
3446 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
3447   \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
3448   % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
3449   % the title of a whole document, so override those back to bold.
3450   \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
3451   % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
3452   % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
3453   \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
3454   \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
3455   \DeclareFieldFormat[booktitle]{\mkbibbold{#1}}
3456   \DeclareFieldFormat[maintitle]{\mkbibbold{#1}}
3457
3458   % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
3459   \DeclareFieldFormat{pagetotal}{#1\addnbspace p\adddot}
3460   \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
3461   \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
3462
3463   % --- Periodical volume / number prefixes ---
3464   \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
3465   \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
3466
3467   % --- "In:" before host title (incollection/inproceedings/inbook) ---
3468   \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
3469
3470   % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
3471   \DeclareFieldFormat{url}{\bibstring{availablefrom}\addcolon\space\url{#1}}
3472
3473   % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
3474   % Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
3475   % differ: the newspaper name (journaltitle) and city, then either
3476   % "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
3477   % "p. <pages>, <date>" (page before date, when none),
3478   % per NBR 6023 §4.2.3.6. ---
3479   \DeclareBibliographyDriver{article}{%

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```

3480 \usebibmacro{bibindex}%
3481 \usebibmacro{begentry}%
3482 \usebibmacro{author/translator+others}%
3483 \setunit{\labelnamepunct}\newblock
3484 \usebibmacro{title}%
3485 \newunit\newblock
3486 \printfield{journalttitle}%
3487 \setunit{\addcomma\space}\printlist{location}%
3488 \iffieldequalstr{entrysubtype}{newspaper}
3489   {\iffieldundef{caderno}
3490     {\setunit{\addcomma\space}\printfield{pages}%
3491       \setunit{\addcomma\space}\printdate}%
3492     {\setunit{\addcomma\space}\printdate%
3493       \setunit{\adddot\space}\printfield{caderno}%
3494       \setunit{\addcomma\space}\printfield{pages}}}%
3495   {\setunit{\addcomma\space}\printfield{volume}%
3496     \setunit{\addcomma\space}\printfield{number}%
3497     \setunit{\addcomma\space}\printfield{pages}%
3498     \setunit{\addcomma\space}\printdate}%
3499 \newunit\newblock
3500 \printfield{doi}%
3501 \newunit\newblock
3502 \usebibmacro{url+urldate}%
3503 \usebibmacro{finentry}}
3504
3505 % --- incollection: ABNT puts the host organizer before the booktitle, as
3506 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
3507 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
3508 \newbibmacro*{coppe:bookeditor}{%
3509   \ifnameundef{editor}
3510     {}
3511     {\printnames{editor}%
3512       \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3513       \clearname{editor}%
3514       \setunit{\adddot\space}}
3515 \newbibmacro*{coppe:partdriver}{%
3516   \usebibmacro{bibindex}\usebibmacro{begentry}%
3517   \usebibmacro{author/translator+others}%
3518   \setunit{\labelnamepunct}\newblock
3519   \usebibmacro{title}%
3520   \newunit\newblock
3521   \usebibmacro{in:}%
3522   \usebibmacro{coppe:bookeditor}%
3523   \printfield{booktitle}%
3524   \newunit\newblock
3525   \printfield{edition}%
3526   \newunit\newblock
3527   \printlist{location}%
3528   \setunit*{\addcolon\space}\printlist{publisher}%
3529   \setunit*{\addcomma\space}\printdate%
3530   \newunit\newblock
3531   \printfield{pages}%
3532   \setunit{\addcomma\space}\printfield{chapter}%
3533   \newunit\newblock

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```

3534 \usebibmacro{url+urldate}%
3535 \usebibmacro{finentry}}
3536 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
3537 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
3538
3539 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
3540 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
3541
3542 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
3543 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}
3544 \DeclareBibliographyDriver{thesis}{%
3545 \usebibmacro{bibindex}\usebibmacro{begentry}%
3546 \usebibmacro{author}%
3547 \setunit{\labelnamepunct}\newblock
3548 \usebibmacro{title}%
3549 \newunit\newblock
3550 \printfield{type}\setunit*{\addspace}\printfield{course}%
3551 \newunit\newblock
3552 \printlist{institution}%
3553 \setunit*{\addcomma\space}\printlist{location}%
3554 \setunit*{\addcomma\space}\printdate%
3555 \newunit\newblock
3556 \printfield{pagetotal}%
3557 \newunit\newblock
3558 \printfield{note}%
3559 \newunit\newblock
3560 \usebibmacro{url+urldate}%
3561 \usebibmacro{finentry}}
3562 % --- Games / audiovisual: render the custom fields (artist, platform,
3563 % director, producer, version) the generic misc driver ignores. ---
3564 \newbibmacro*{coppe:media}{%
3565 \iflistundef{artist}{\setunit{\addperiod\space}%
3566 \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
3567 \iffieldundef{platform}{\setunit{\addperiod\space}%
3568 \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%
3569 \iflistundef{director}{\setunit{\addperiod\space}%
3570 \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
3571 \iflistundef{producer}{\setunit{\addperiod\space}%
3572 \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}%
3573 \newbibmacro*{coppe:mediadriv}{%
3574 \usebibmacro{bibindex}\usebibmacro{begentry}%
3575 \usebibmacro{author/translator+others}%
3576 \setunit{\labelnamepunct}\newblock
3577 \usebibmacro{title}%
3578 \newunit\newblock
3579 \usebibmacro{coppe:media}%
3580 \newunit\newblock
3581 \printlist{publisher}%
3582 \setunit*{\addcomma\space}\printlist{location}%
3583 \setunit*{\addcomma\space}\printdate%
3584 \newunit\newblock
3585 \printfield{version}%
3586 \newunit\newblock
3587 \printfield{note}%

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3588 \newunit\newblock
3589 \usebibmacro{url+urldate}%
3590 \usebibmacro{finentry}}
3591 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
3592 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
3593 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
3594 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
3595 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
3596 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
3597 % imprenta) plus the mandatory scale ("Escala 1:N").
3598 \DeclareFieldFormat{scale}{\bibstring{coppescale}\adnbspace#1}
3599 \DeclareBibliographyDriver{map}{%
3600 \usebibmacro{bibindex}\usebibmacro{begentry}%
3601 \usebibmacro{author/translator+others}%
3602 \setunit{\labelnamepunct}\newblock
3603 \usebibmacro{title}%
3604 \newunit\newblock
3605 \printlist{location}%
3606 \setunit*{\addcolon\space}\printlist{publisher}%
3607 \setunit*{\addcomma\space}\printdate%
3608 \newunit\newblock
3609 \printfield{scale}%
3610 \newunit\newblock
3611 \usebibmacro{url+urldate}%
3612 \usebibmacro{finentry}}
3613 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
3614 % number. date. The holder (depositante) is printed in normal case, not the
3615 % ALL-CAPS family treatment the bibliography gives the leading author.
3616 \newbibmacro*{coppe:depositor}{%
3617 \ifnameundef{holder}
3618 {}
3619 {\printtext{\bibstring{depositor}\addcolon\space}%
3620 \begingroup
3621 \def\mkbibnamefamily##1{##1}%
3622 \def\mkbibnameprefix##1{##1}%
3623 \printnames{holder}%
3624 \endgroup}}
3625 \DeclareBibliographyDriver{patent}{%
3626 \usebibmacro{bibindex}%
3627 \usebibmacro{begentry}%
3628 \usebibmacro{author/translator+others}%
3629 \setunit{\labelnamepunct}\newblock
3630 \usebibmacro{title}%
3631 \newunit\newblock
3632 \usebibmacro{coppe:depositor}%
3633 \newunit\newblock
3634 \printfield{type}%
3635 \setunit*{\addspace}\printfield{number}%
3636 \newunit\newblock
3637 \printdate%
3638 \newunit\newblock
3639 \usebibmacro{url+urldate}%
3640 \usebibmacro{finentry}}
3641 % Exotic ABNT types that biblatex's standard style leaves without a driver:

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```

3642 % give each a baseline (book-like for published docs: norma/partitura; misc for
3643 % the rest) so they print. Per-type ABNT refinements can follow later.
3644 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
3645 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
3646 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
3647 % the law id + ementa (plain, not bold), then location and date.
3648 \newbibmacro*{coppe:legaldriver}{%
3649   \usebibmacro{bibindex}\usebibmacro{begentry}%
3650   \usebibmacro{author/translator+others}%
3651   \setunit{\labelnamepunct}\newblock
3652   \usebibmacro{title}%
3653   \newunit\newblock
3654   % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
3655   \iffielddundef{relator}{}\{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\newblock\}
3656   \printlist{location}%
3657   \setunit{\addcomma\space}\printdate%
3658   \newunit\newblock
3659   \usebibmacro{url+urldate}%
3660   \usebibmacro{finentry}}
3661 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
3662 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
3663 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
3664 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
3665 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
3666 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
3667 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
3668 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
3669 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
3670 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
3671
3672 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
3673 \DeclareSourcemap{%
3674   \maps[datatype=bibtex]{%
3675     % entry-type PT synonyms -> canonical English
3676     \map{step[typesource=livro,typetarget=book]}
3677     \map{step[typesource=artigo,typetarget=article]}
3678     \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtype,typetarget=article]}
3679     \map{step[typesource=partedelivro,typetarget=incollection]}
3680     \map{step[typesource=emlivro,typetarget=inbook]}
3681     \map{step[typesource=verbete,typetarget=inreference]}
3682     \map{step[typesource=anais,typetarget=proceedings]}
3683     \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
3684     \map{step[typesource=periodico,typetarget=periodical]}
3685     \map{step[typesource=relatorio,typetarget=report]}
3686     \map{step[typesource=relatoriotecnico,typetarget=report]}
3687     \map{step[typesource>manual,typetarget>manual]}
3688     \map{step[typesource=correspondencia,typetarget=letter]}
3689     \map{step[typesource=patente,typetarget=patent]}
3690     \map{step[typesource=diverso,typetarget=misc]}
3691     \map{step[typesource=jogo,typetarget=game]}
3692     \map{step[typesource=jogoeletronico,typetarget=videogame]}
3693     \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
3694     \map{step[typesource=filme,typetarget=movie]}
3695     \map{step[typesource=serietv,typetarget=tvshow]}

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3696 \map{\step[typesource=programadetelevision,typetarget=tvshow]}
3697 % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
3698 % except map (declared in coppe.dbx).
3699 \map{\step[typesource=legislacao,typetarget=legislation]}
3700 \map{\step[typesource=atonormativo,typetarget=legislation]}
3701 \map{\step[typesource=jurisprudencia,typetarget=jurisdiction]}
3702 \map{\step[typesource=documentojuridico,typetarget=legal]}
3703 \map{\step[typesource=documentocivil,typetarget=legal]}
3704 \map{\step[typesource=norma,typetarget=standard]}
3705 \map{\step[typesource=partitura,typetarget=music]}
3706 \map{\step[typesource=iconografico,typetarget=image]}
3707 \map{\step[typesource=obradearte,typetarget=artwork]}
3708 \map{\step[typesource=tridimensional,typetarget=artwork]}
3709 \map{\step[typesource=cartografico,typetarget=map]}
3710 \map{\step[typesource=mapa,typetarget=map]}
3711 \map{\step[typesource=audiovisual,typetarget=video]}
3712 \map{\step[typesource=conjuntodedados,typetarget=dataset]}
3713 \map{\step[typesource=fasciculo,typetarget=supperperiodical]}
3714 \map{\step[typesource=trabalhoapresentado,typetarget=unpublished]}
3715 % thesis family: retype to thesis and preset the (babel-aware) type word
3716 \map{\step[typesource=tese,typetarget=thesis]}
3717 \map{\step[typesource=dissertacao,typetarget=thesis]}
3718 \map{\step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fieldvalue=masterdissertation]}
3719 \map{\step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fieldvalue=mscdissertation]}
3720 \map{\step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fieldvalue=dissertacaomestrado]}
3721 \map{\step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=dscthesis]}
3722 \map{\step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tesedoutorado]}
3723 \map{\step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phdthesis]}
3724 \map{\step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tesephd]}
3725 \map{\step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
3726 % field PT synonyms (unaccented) -> canonical English
3727 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
3728 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
3729 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
3730 \map{\step[fieldsource=titulo,fieldtarget=title]}
3731 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
3732 \map{\step[fieldsource=autor,fieldtarget=author]}
3733 \map{\step[fieldsource=editora,fieldtarget=publisher]}
3734 \map{\step[fieldsource=local,fieldtarget=location]}
3735 \map{\step[fieldsource=ano,fieldtarget=year]}
3736 \map{\step[fieldsource=data,fieldtarget=date]}
3737 \map{\step[fieldsource=edicao,fieldtarget=edition]}
3738 \map{\step[fieldsource=paginas,fieldtarget=pages]}
3739 \map{\step[fieldsource=numero,fieldtarget=number]}
3740 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
3741 \map{\step[fieldsource=organizador,fieldtarget=editor]}
3742 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
3743 \map{\step[fieldsource=tipo,fieldtarget=type]}
3744 \map{\step[fieldsource=nota,fieldtarget=note]}
3745 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
3746 \map{\step[fieldsource=curso,fieldtarget=course]}
3747 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
3748 \map{\step[fieldsource=artista,fieldtarget=artist]}
3749 \map{\step[fieldsource=plataforma,fieldtarget=platform]}

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3750 \map{\step[fieldsource=diretor,fieldtarget=director]}
3751 \map{\step[fieldsource=produtor,fieldtarget=producer]}
3752 \map{\step[fieldsource=versao,fieldtarget=version]}
3753 \map{\step[fieldsource=escala,fieldtarget=scale]}
3754 }%
3755 }
3756
3757 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
3758 \setlength{\bibhang}{0pt}
3759 \setlength{\bibitemsep}{\baselineskip}
3760
3761 </bbx>
3762 <*cbx>
3763 \ProvidesFile{coppe.cbx}[2026/05/28 v4.0 CoppeTeX ABNT author-date citations]
3764 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
3765 % author-year cites with normal-caps names (post-2020 rule) out of the box.
3766 \RequireCitationStyle{authoryear-comp}
3767
3768 % --- NBR 10520:2023 author separator -----
3769 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
3770 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
3771 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
3772 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
3773 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
3774 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
3775 \DeclareDelimFormat{textcite}{\multinamedelim}{\addcomma\space}
3776 \DeclareDelimFormat{textcite}{finalnamedelim}{\addspace\bibstring{and}\space}
3777
3778 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
3779 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
3780 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
3781 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
3782 % so it is never added to the bibliography.
3783 \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3784 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
3785 \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
3786 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
3787 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
3788 \DeclareRobustCommand{\citetapud}[4][]{%
3789 #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
3790 \DeclareRobustCommand{\citepapud}[4][]{%
3791 \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
3792 \citeauthor{#4}\addcomma\space\citeyear{#4}}%
3793 \ifblank{#1}{ }\addcomma\space\mkpageprefix[pagination]{#1}}}%
3794 </cbx>
3795 <*lbxbr>
3796 \ProvidesFile{brazilian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX brazilian strings]
3797 \InheritBibliographyExtras{brazilian}
3798 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
3799 \DeclareBibliographyStrings{%
3800 inherit = {brazilian},
3801 availablefrom = {{Dispon}'{\i}vel em}{Dispon}'{\i}vel em}},
3802 urlseen = {{Acesso em}{Acesso em}},
3803 in = {{In}{In}},

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3804 editor      = {{organizador}{org\adddot}},
3805 editors      = {{organizadores}{org\adddot}},
3806 depositor    = {{Depositante}{Depositante}},
3807 coppescale   = {{Escala}{Escala}},
3808 msdiss       = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
3809 dscthesi     = {{Tese de Doutorado}{Tese de Doutorado}},
3810 phdthesis    = {{Tese de Doutorado}{Tese de Doutorado}},
3811 tcc          = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}},
3812 }
3813 </lbxbr>
3814 <*lbxen>
3815 \ProvidesFile{english-coppe.lbx}[2026/05/28 v4.0 CoppeTeX english strings]
3816 \InheritBibliographyExtras{english}
3817 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
3818 \DeclareBibliographyStrings{%
3819   inherit      = {english},
3820   availablefrom = {{Available from}{Available from}},
3821   urlseen      = {{Access on}{Access on}},
3822   in           = {{In}{In}},
3823   editor       = {{editor}{ed\adddot}},
3824   editors      = {{editors}{ed\adddot}},
3825   depositor    = {{Depositor}{Depositor}},
3826   coppescale   = {{Scale}{Scale}},
3827   msdiss       = {{M\adddot Sc\adddot\ Dissertation}{M\adddot Sc\adddot\ Diss\adddot}},
3828   dscthesi     = {{D\adddot Sc\adddot\ Thesis}{D\adddot Sc\adddot\ Thesis}},
3829   phdthesis    = {{Ph\adddot D\adddot\ Thesis}{Ph\adddot D\adddot\ Thesis}},
3830   tcc          = {{Undergraduate Thesis}{Undergraduate Thesis}},
3831 }
3832 </lbxen>
3833 <*lbxes>
3834 \ProvidesFile{spanish-coppe.lbx}[2026/05/28 v4.0 CoppeTeX spanish strings]
3835 \InheritBibliographyExtras{spanish}
3836 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
3837 \DeclareBibliographyStrings{%
3838   inherit      = {spanish},
3839   availablefrom = {{Disponible en}{Disponible en}},
3840   urlseen      = {{Consultado el}{Consultado el}},
3841   in           = {{En}{En}},
3842   editor       = {{editor}{ed\adddot}},
3843   editors      = {{editores}{ed\adddot}},
3844   depositor    = {{Solicitante}{Solicitante}},
3845   coppescale   = {{Escala}{Escala}},
3846   msdiss       = {{Tesis de Maestr'\i{a}}{Tesis de Maestr'\i{a}}},
3847   dscthesi     = {{Tesis Doctoral}{Tesis Doctoral}},
3848   phdthesis    = {{Tesis Doctoral}{Tesis Doctoral}},
3849   tcc          = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
3850 }
3851 </lbxes>
3852 <*lbxfr>
3853 \ProvidesFile{french-coppe.lbx}[2026/05/28 v4.0 CoppeTeX french strings]
3854 \InheritBibliographyExtras{french}
3855 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
3856 \DeclareBibliographyStrings{%
3857   inherit      = {french},

```

```

3858 availablefrom = {{Disponibile \'a}{Disponibile \'a}},
3859 urlseen       = {{Consult\'e le}{Consult\'e le}},
3860 in            = {{Dans}{Dans}},
3861 editor        = {{\'editeur}{\'ed\addot}},
3862 editors       = {{\'editeurs}{\'ed\addot}},
3863 depositor     = {{D\'eposant}{D\'eposant}},
3864 coppescale    = {{\'Echelle}{\'Echelle}},
3865 msdiss        = {{M\'emoire de Master}{M\'emoire de Master}},
3866 dscthesi      = {{Th\'ese de Doctorat}{Th\'ese de Doctorat}},
3867 phdthesis     = {{Th\'ese de Doctorat}{Th\'ese de Doctorat}},
3868 tcc           = {{M\'emoire de fin d\'\'etudes}{M\'emoire de fin d\'\'etudes}},
3869 }
3870 </lboxfr>
3871 <*lboxit>
3872 \ProvidesFile{italian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX italian strings]
3873 \InheritBibliographyExtras{italian}
3874 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
3875 \DeclareBibliographyStrings{%
3876   inherit      = {italian},
3877   availablefrom = {{Disponibile da}{Disponibile da}},
3878   urlseen      = {{Consultato il}{Consultato il}},
3879   in           = {{In}{In}},
3880   editor       = {{curatore}{cur\addot}},
3881   editors      = {{curatori}{cur\addot}},
3882   depositor    = {{Depositante}{Depositante}},
3883   coppescale   = {{Scala}{Scala}},
3884   msdiss       = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
3885   dscthesi     = {{Tesi di Dottorato}{Tesi di Dottorato}},
3886   phdthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
3887   tcc          = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
3888 }
3889 </lboxit>
3890 <*langes>
3891 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
3892 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
3893 %% spanish slot) is active. Populates the dispatcher table for the
3894 %% spanish language and registers the biblalex language mapping.
3895 %% Institutional names (universityname/cityname/statename/countryname)
3896 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
3897 %% rosto are a Brazilian institutional template that does not translate.
3898 \ProvidesFile{coppe-lang-spanish.def}[2026/05/28 v4.0 CoppeTeX spanish class strings]
3899 \copperdefstring{spanish}{appendix}    {Ap\'endice}
3900 \copperdefstring{spanish}{annex}       {Anexo}
3901 \copperdefstring{spanish}{frame}       {Cuadro}
3902 \copperdefstring{spanish}{listframe}    {Lista de Cuadros}
3903 \copperdefstring{spanish}{source}       {Fuente}
3904 \copperdefstring{spanish}{program}      {Programa}
3905 \copperdefstring{spanish}{listprogram}  {Lista de Programas}
3906 \copperdefstring{spanish}{references}   {Referencias}
3907 \copperdefstring{spanish}{in:}         {en}
3908 \copperdefstring{spanish}{art}         {Arte}
3909 \copperdefstring{spanish}{platform}     {Plataforma}
3910 \copperdefstring{spanish}{directedby}   {Direcci\'on}
3911 \copperdefstring{spanish}{producedby}   {Producci\'on}

```

```

3912 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
3913 \copperdefstring{spanish}{listsymbol}      {Lista de S\'imbolos}
3914 \copperdefstring{spanish}{glossary}         {Glosario}
3915 \copperdefstring{spanish}{advisor}          {Director}
3916 \copperdefstring{spanish}{advisors}         {Directores}
3917 \copperdefstring{spanish}{dept}             {Programa}
3918 \copperdefstring{spanish}{approvedmale}     {Aprobado por}
3919 \copperdefstring{spanish}{approvedfemale}   {Aprobada por}
3920 \copperdefstring{spanish}{approvedonmale}   {Aprobado el}
3921 \copperdefstring{spanish}{approvedonfemale}{Aprobada el}
3922 \copperdefstring{spanish}{keywordsname}     {Palabras clave}
3923 \copperdefstring{spanish}{volumename}       {Volumen}
3924 \copperdefstring{spanish}{coadvisor}        {Codirector}
3925 \copperdefstring{spanish}{coadvisors}       {Codirectores}
3926 \copperdefstring{spanish}{ofname}          {de}
3927 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
3928 \copperdefstring{spanish}{cityname}        {Rio de Janeiro}
3929 \copperdefstring{spanish}{statename}        {RJ}
3930 \copperdefstring{spanish}{countryname}     {Brasil}
3931 \copperdefstring{spanish}{monthname1}       {enero}
3932 \copperdefstring{spanish}{monthname2}       {febrero}
3933 \copperdefstring{spanish}{monthname3}       {marzo}
3934 \copperdefstring{spanish}{monthname4}       {abril}
3935 \copperdefstring{spanish}{monthname5}       {mayo}
3936 \copperdefstring{spanish}{monthname6}       {junio}
3937 \copperdefstring{spanish}{monthname7}       {julio}
3938 \copperdefstring{spanish}{monthname8}       {agosto}
3939 \copperdefstring{spanish}{monthname9}       {septiembre}
3940 \copperdefstring{spanish}{monthname10}      {octubre}
3941 \copperdefstring{spanish}{monthname11}      {noviembre}
3942 \copperdefstring{spanish}{monthname12}     {diciembre}
3943 \copperdefstring{spanish}{algorithm2eopt}{spanish}
3944 \copperdefstring{spanish}{babelname}       {spanish}
3945 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
3946 %% language-mapping registration until \begin{document} so biblatex's
3947 %% \DeclareLanguageMapping is available.
3948 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
3949 \end{langes}
3950 \end{langfr}
3951 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
3952 %% See coppe-lang-spanish.def for the structure. Institutional names
3953 %% remain Portuguese; only document-content strings translate.
3954 \ProvidesFile{coppe-lang-french.def}[2026/05/28 v4.0 CoppeTeX french class strings]
3955 \copperdefstring{french}{appendix}           {Annexe}
3956 \copperdefstring{french}{annex}             {Annexe}
3957 \copperdefstring{french}{frame}             {Encadr\'e}
3958 \copperdefstring{french}{listframe}         {Liste des encadr\'es}
3959 \copperdefstring{french}{source}            {Source}
3960 \copperdefstring{french}{program}           {Programme}
3961 \copperdefstring{french}{listprogram}       {Liste des programmes}
3962 \copperdefstring{french}{references}        {R\'ef\'erences}
3963 \copperdefstring{french}{in:}              {dans}
3964 \copperdefstring{french}{art}              {Illustration}
3965 \copperdefstring{french}{platform}         {Plateforme}

```

```

3966 \copperdefstring{french}{directedby}      {R\realis\’e par}
3967 \copperdefstring{french}{producedby}      {Produit par}
3968 \copperdefstring{french}{listabbreviation}{Liste des abr\’eviations et acronymes}
3969 \copperdefstring{french}{listsymbol}       {Liste des symboles}
3970 \copperdefstring{french}{glossary}         {Glossaire}
3971 \copperdefstring{french}{advisor}          {Directeur}
3972 \copperdefstring{french}{advisors}         {Directeurs}
3973 \copperdefstring{french}{dept}             {Programme}
3974 \copperdefstring{french}{approvedmale}     {Approuv\’e par}
3975 \copperdefstring{french}{approvedfemale}   {Approuv\’ee par}
3976 \copperdefstring{french}{approvedonmale}   {Approuv\’e le}
3977 \copperdefstring{french}{approvedonfemale}{Approuv\’ee le}
3978 \copperdefstring{french}{keywordsname}     {Mots-cl\’es}
3979 \copperdefstring{french}{volumename}       {Volume}
3980 \copperdefstring{french}{coadvisor}        {Co-directeur}
3981 \copperdefstring{french}{coadvisors}       {Co-directeurs}
3982 \copperdefstring{french}{ofname}           {de}
3983 \copperdefstring{french}{universityname}{Universidade Federal do Rio de Janeiro}
3984 \copperdefstring{french}{cityname}         {Rio de Janeiro}
3985 \copperdefstring{french}{statename}        {RJ}
3986 \copperdefstring{french}{countryname}      {Brasil}
3987 \copperdefstring{french}{monthname1}       {janvier}
3988 \copperdefstring{french}{monthname2}      {f\’evrier}
3989 \copperdefstring{french}{monthname3}       {mars}
3990 \copperdefstring{french}{monthname4}       {avril}
3991 \copperdefstring{french}{monthname5}       {mai}
3992 \copperdefstring{french}{monthname6}      {juin}
3993 \copperdefstring{french}{monthname7}      {juillet}
3994 \copperdefstring{french}{monthname8}      {ao\’ut}
3995 \copperdefstring{french}{monthname9}      {septembre}
3996 \copperdefstring{french}{monthname10}{octobre}
3997 \copperdefstring{french}{monthname11}{novembre}
3998 \copperdefstring{french}{monthname12}{d\’ecembre}
3999 \copperdefstring{french}{algorithm2eopt}{french}
4000 \copperdefstring{french}{babelname}       {french}
4001 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
4002 \</langfr>
4003 \*langit)
4004 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
4005 %% See coppe-lang-spanish.def for the structure. The algorithm2e
4006 %% language option for Italian is spelled "italiano" -- that’s the
4007 %% spelling algorithm2e ships, even though babel calls the language
4008 %% "italian".
4009 \ProvidesFile{coppe-lang-italian.def}[2026/05/28 v4.0 CoppeTeX italian class strings]
4010 \copperdefstring{italian}{appendix}         {Appendice}
4011 \copperdefstring{italian}{annex}            {Allegato}
4012 \copperdefstring{italian}{frame}            {Riquadro}
4013 \copperdefstring{italian}{listframe}        {Elenco dei riquadri}
4014 \copperdefstring{italian}{source}           {Fonte}
4015 \copperdefstring{italian}{program}          {Programma}
4016 \copperdefstring{italian}{listprogram}      {Elenco dei programmi}
4017 \copperdefstring{italian}{references}       {Riferimenti}
4018 \copperdefstring{italian}{in:}              {in}
4019 \copperdefstring{italian}{art}              {Illustrazione}

```

```

4020 \copperdefstring{italian}{platform}      {Piattaforma}
4021 \copperdefstring{italian}{directedby}     {Regia di}
4022 \copperdefstring{italian}{producedby}     {Prodotto da}
4023 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
4024 \copperdefstring{italian}{listsymbol}     {Elenco dei simboli}
4025 \copperdefstring{italian}{glossary}       {Glossario}
4026 \copperdefstring{italian}{advisor}       {Relatore}
4027 \copperdefstring{italian}{advisors}      {Relatori}
4028 \copperdefstring{italian}{dept}          {Programma}
4029 \copperdefstring{italian}{approvedmale}   {Approvato da}
4030 \copperdefstring{italian}{approvedfemale} {Approvata da}
4031 \copperdefstring{italian}{approvedonmale} {Approvato il}
4032 \copperdefstring{italian}{approvedonfemale}{Approvata il}
4033 \copperdefstring{italian}{keywordsname} {Parole chiave}
4034 \copperdefstring{italian}{volumename} {Volume}
4035 \copperdefstring{italian}{coadvisor} {Correlatore}
4036 \copperdefstring{italian}{coadvisors} {Correlatori}
4037 \copperdefstring{italian}{ofname} {di}
4038 \copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
4039 \copperdefstring{italian}{cityname} {Rio de Janeiro}
4040 \copperdefstring{italian}{statename} {RJ}
4041 \copperdefstring{italian}{countryname} {Brasil}
4042 \copperdefstring{italian}{monthname1} {gennaio}
4043 \copperdefstring{italian}{monthname2} {febbraio}
4044 \copperdefstring{italian}{monthname3} {marzo}
4045 \copperdefstring{italian}{monthname4} {aprile}
4046 \copperdefstring{italian}{monthname5} {maggio}
4047 \copperdefstring{italian}{monthname6} {giugno}
4048 \copperdefstring{italian}{monthname7} {luglio}
4049 \copperdefstring{italian}{monthname8} {agosto}
4050 \copperdefstring{italian}{monthname9} {settembre}
4051 \copperdefstring{italian}{monthname10}{ottobre}
4052 \copperdefstring{italian}{monthname11}{novembre}
4053 \copperdefstring{italian}{monthname12}{dicembre}
4054 \copperdefstring{italian}{algorithm2eopt}{italiano}
4055 \copperdefstring{italian}{babelname} {italian}
4056 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
4057 \end{langit}
4058 \newcommand*{numbbx}{}
4059 \ProvidesFile{coppe-numeric.bbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric bibliography]
4060 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
4061 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
4062 % the PT->EN source map and the language mappings) and only adds the [n] list
4063 % label and the numeric sorting on top.
4064 \RequireBibliographyStyle{coppe}
4065
4066 % ABNT numeric system lists references in the order they are first cited
4067 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
4068 % numbered list. 'labelnumber' generates the [n] field.
4069 \ExecuteBibliographyOptions{labelnumber,sorting=none}
4070
4071 % Bracketed labels, both in the list and for any shorthand entries.
4072 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
4073 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}

```

```

4074
4075 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
4076 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
4077 \defbibenvironment{bibliography}
4078   {\list
4079     {\printtext[labelnumberwidth]{%
4080       \printfield{labelprefix}%
4081       \printfield{labelnumber}}}
4082     {\setlength{\labelwidth}{\labelnumberwidth}%
4083      \setlength{\leftmargin}{\labelwidth}%
4084      \setlength{\labelsep}{\biblabelsep}%
4085      \addtolength{\leftmargin}{\labelsep}%
4086      \setlength{\itemsep}{\bibitemsep}%
4087      \setlength{\parsep}{\bibparsep}}%
4088     \renewcommand*{\makelabel}[1]{\hss##1}}
4089   {\endlist}
4090   {\item}
4091
4092 </numbbx>
4093 <*numcbx>
4094 \ProvidesFile{coppe-numeric.cbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric citations]
4095 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
4096 % numbers [1], compressed and sorted within a single cite group. This is the
4097 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
4098 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
4099 % \citet -> Author [1].
4100 \RequireCitationStyle{numeric-comp}
4101 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

10.3 The generated companion files

Everything the distribution ships is extracted from this file: the class, the BibLaTeX styles, the language packs, the databases, the `.ist`, the demonstration documents and the `latexmkrc`. One `pdflatex coppe.ins` rebuilds all of it and there is no second place to edit. Each one sits in an `\iffalse` block: the documentation driver never typesets them, and only `coppe.ins` extracts them.

The line is drawn at what is DISTRIBUTED. What exists only to prove the class works stays outside, as ordinary files under version control:

`tests/` the regression suite. Nobody installing CoppeTeX needs it, and generating it from here would put proof and product in the same box.

`adversativa/` twelve documents – four kinds of work by three languages – each firing everything the class offers at once.

`tools/*.ps1` the build harness itself.

`NORMA_COPPE_2026.tex`, `futuremanual2026.tex` documents about the norm that live in this repository with their own life cycle.

Both suites are run by one command, `tools\prova.ps1`, and that is what has to come out clean before a version is tagged.

10.3.1 `example_pt.tex` -- pt-main demo

`example_pt.tex` – CoppeTeX 4.0 multilingual demo, main language: brazilian. Short demonstration document (cover + folha + abstracts + 1 chapter) used by the CPGP submission package. The full pt-main example with bibliography and lists is the historical `example.tex`.

10.3.2 `example_en.tex` -- en-main demo

`example_en.tex` – CoppeTeX 4.0 multilingual demo, main language: english.

10.3.3 `example_es.tex` -- es-main demo

`example_es.tex` – CoppeTeX 4.0 multilingual demo, main language: spanish. Requires the language pack files `coppe-lang-spanish.def` and `spanish-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.4 `example_fr.tex` -- fr-main demo

`example_fr.tex` – CoppeTeX 4.0 multilingual demo, main language: french. Requires the language pack files `coppe-lang-french.def` and `french-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.5 `example_it.tex` -- it-main demo

`example_it.tex` – CoppeTeX 4.0 multilingual demo, main language: italian. Requires the language pack files `coppe-lang-italian.def` and `italian-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.6 `example_pdfa.tex` -- the full example as PDF/A

`example_pdfa.tex` – o `example.tex` completo, compilado com a opção ‘pdfa’.

Existe para dar ao veraPDF o documento mais exigente que a classe produz: o exemplo inteiro, com bibliografia, listas, figuras, tabelas, algoritmos e as caixas do `tcolorbox` – ou seja, transparência, que é justamente o que o a-1b proibiria e o a-2b admite. Validar só um documento curto não diria nada sobre o que vai de fato para o depósito.

`\PassOptionsToClass` tem de vir ANTES do `\documentclass` que o `example.tex` traz dentro de si; é o único jeito de acrescentar uma opção de classe a um arquivo gerado pelo docstrip sem duplicá-lo aqui e deixar as duas cópias divergirem com o tempo.

10.3.7 covers_5languages.tex -- the five covers side by side

`covers_5languages.tex` – One-page visual demonstration of the CoppeTeX 4.0 multilingual model: the five built-in main languages side by side, all sharing the same Portuguese institutional template, differing only in the title (and the body language, not visible on the cover).

This file uses the article class deliberately – it is a meta-document about the coppe class, not a thesis. It expects the five cover PNGs `cover_pt.png`, `cover_en.png`, `cover_es.png`, `cover_fr.png`, `cover_it.png` to live in the same directory (generated by `pdftoppm` from the `example_<lang>.pdf` demos).

10.3.8 latexmkrc -- latexmk/Overleaf build configuration

`latexmkrc` – CoppeTeX build configuration (works locally and on Overleaf).

Run with: `latexmk example` (or `latexmk futuremanual2026`, your thesis...) `latexmk` reads this file automatically when it is in the build directory. On Overleaf, place this file at the project root; Overleaf runs `latexmk`.

Engine: pdfLaTeX by default (fast). For LuaLaTeX (recommended, full Unicode) set `$pdf_mode = 4` below, or use the `"% !TeX program = lualatex"` magic comment / the Overleaf "Menu > Compiler" setting.

Acknowledgments

Thanks to all COPPET_EX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

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Numbers written in *italic* refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in *roman* refer to the code lines where the entry is used.

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Change History

v0.0		v3.1	
General: Creation Date.	1	<code>\department</code> : Included a sort key in <code>symb</code>	52
v0.1		v3.2	
General: Documentation:		General: Fixed version problem between <code>cls</code> and <code>dtx</code>	1
bibliography fixed, title translation.	1	v3.3	
Sourceforge submission.	1	General: Extend <code>abrev</code> to work like <code>symb</code> and accept a sorting key .	1
v0.2		v3.4	
General: Unification of the code for the list of symbols and abbreviations.	1	General: Some examples for figures, tables, longtables, etc. .	1
v0.3		v3.4.1	
General: Added 'draft' option. . . .	1	General: Fix english option to consider brazilian a second language, describe it in document	51
<code>\maketitle</code> : Added number of examiners test.	61	v3.5	
Generalization.	61	General: New command <code>annex</code> . .	97
v0.4		v3.5.1	
General: Beta documentation. . . .	1	General: New command <code>annex</code> now supports brazilian or english .	97
v0.5		v4.0	
<code>abstract</code> : Changed from macro to environment.	68	General: Catalog-label writers (<code>coppe@mainBegin</code> , <code>coppe@bibBegin</code> , <code>coppe@bibEnd</code> , <code>coppe@hasLof</code>) made language-agnostic: <code>detokenize</code> wrap on label names protects ':' and '.' from babel shorthand activation, and the raw <code>etex</code> primitives <code>romannumeral</code> / <code>number</code> replace LaTeX's <code>roman</code> / <code>arabic</code> to bypass <code>spanish-babel</code> 's <code>small-caps @roman</code> redefinition that would otherwise crash <code>pdftex</code> with "Unbalanced write command" on every <code>es-main</code> document.	1
<code>\backmatter</code> : Added <code>mainmatter</code> pages counter.	57	Class-wide string dispatcher (<code>copperdefstring</code> , <code>coppestring</code> , <code>coppemainstring</code> , <code>coppeforeignstring</code>) replaces scattered <code>iflanguage</code> and <code>if</code> english branches; <code>pt/en</code> outputs remain byte-identical. .	1
<code>foreignabstract</code> : Changed from macro to environment.	69	Multilingual model: three fixed slots (<code>main</code> , <code>foreign</code> , <code>optional</code> <code>third</code>); five built-in language options (<code>brazilian</code> , <code>english</code> ,	
v1.0			
General: First COPPETeX release. .	1		
v2.0			
General: COPPETeX release 2.0. . .	1		
v2.1			
General: COPPETeX release 2.1:			
Matching the new rules.	1		
<code>\advisor</code> : Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors. .	54		
v2.2			
General: Matching new guidelines, including new logo.	1		
v2.2.2			
General: Fixed some text constants in <code>.bib</code> and documented it here. .	1		
v3.0			
General: Added support for monographs in English.	51		
New approval page layout. . . .	63		
<code>\department</code> : Added new course on Nanotechnology.	52		
<code>\examiner</code> : Examiners expansion without degree.	54		

spanish, french, italian);	usecoppelanguage,
plug-in mechanism for any	brazilianabstract environment. . . 1
other Babel language via	
coppe-lang-lang.def plus	\coppetexfinalpage: Initial
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